

# Finesse 3

## Through the Looking Glass

Miron van der Kolk, Jeremie Gobeil,
Daniel Brown and Andreas Freise

# High level overview

- Who are Jeremie and Miron (and Mischa?)
- Into to Finesse from developer POV – Code overview (Python/Cython/C)
- Plane-wave single mirror
- HOMs → cavity
- Multiple frequencies → Modulator
- Signal modelling
- Sparse matrix
- Using Xaxis to change the power of the laser
- More actions ⇒ Minimizer/Lock
- Custom Actions/Components/Detector

Questions to answer

- How Finesse go from a model to a matrix?
- How Finesse solve the matrix?
- Why we are using sparse matrices?
- How do detectors work?
- What is the consequence of HOMs on the matrices?
- How to extend the code? => Workspace?
- (What are the actions?) (maybe show a minimizer?)
- How to create and what information is important for an issue?

# Finesse

## Software Engineers Team







| | Jeremie | Miron | Mischa |
|---|---|---|---|
| coding | ||||||||||| | ||||||||||| | ||||||||||| |
| python fu | | |

# Finesse : the developer point-of-view

# $ ls --pov=user ~/finesse3

A Python library

KatScript

```
import finesse
from finesse.analysis.actions import Xaxis
finesse.init_plotting()

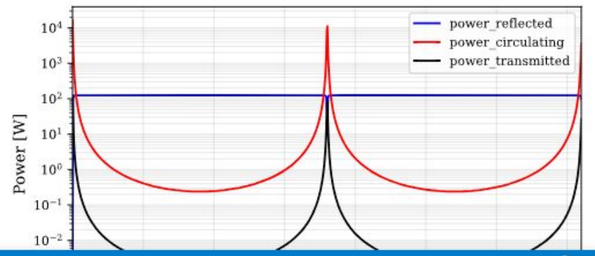
OMC_model = finesse.Model()
OMC_model.parse(
    """
    l l1 P=125                  # laser with P=125W
    s s1 l1.p1 OMC_1.p1 L=1     # space of 1m length connectint laser to OMC

    bs OMC_1    T=0.0076    L=0
    bs OMC_2    T=0.0075    L=0
    bs OMC_3    T=36.0u     L=0
    bs OMC_4    T=35.9u     L=0

    # Connect spaces to form a bow tie
    s sOMC_1    OMC_1.p3    OMC_2.p1    L=0.2815
    s sOMC_2    OMC_2.p2    OMC_3.p1    L=0.2842
    s sOMC_3    OMC_3.p2    OMC_4.p1    L=0.2815
    s sOMC_4    OMC_4.p2    OMC_1.p4    L=0.2842

    # Photodiodes measuring the reflected, circulating, and transmitted light
    pd power_reflected      OMC_1.p2.o
    pd power_circulating    OMC_1.p3.o
    pd power_transmitted    OMC_2.p3.o
    """
)

# Change the tuning of mirror OMC_2 from 0 to 360 degrees
out = OMC_model.run(Xaxis('OMC_2.phi', 'lin', 0, 360, 1000))
out.plot(logy=True);
```




Documentation

Finesse

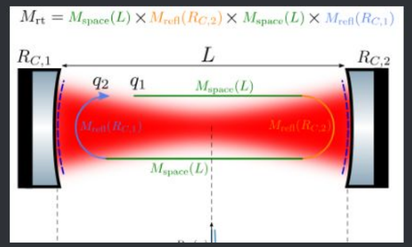
Search docs

- What is Finesse
- More about Finesse
- Installing Finesse
- Getting started
- Examples
- User guide
  - Plane-wave optics
  - Building and interacting with models
  - Frequency-domain modelling
  - Higher-order Hermite-Gauss modes
    - Phase configuration settings
    - Transformation of the beam parameter
    - Cavity eigenmodes
    - Tracing the beam

where $A, B, C$ and $D$ are the element
beam is recreated after each round-trip
compute the parameter $q_{cav} \equiv q_1 = q_2$ by solving:

$$Cq_{cav}^2 + (D - A)q_{cav} - B = 0,$$

An example of this is shown in Fig. 15 where the round trip matrix is given at the top of the figure. From this, we can compute the $A, B, C$ and $D$ coefficients for the round-trip matrix to solve the eigenmode equation above. This quadratic equation generally has two solutions, one being the complex conjugate of the other.




Communication channels

Finesse3 users

```
model = finesse.Model("""
l l1
s s1 l1.p1 m1.p1
m m1 T=l1.f*1e-12  # insert the dependence that you need in here
s s2 m1.p2 m2.p1
m m2
""")

model.run(Xaxis(model.l1.f, 'log', f_start, f_stop, steps))
```

2 replies — MarijnS Update: I found the problem. There was a spelling mistake in "...

Today

Miron van der Kolk 09:45

Dear Finesse3 users,

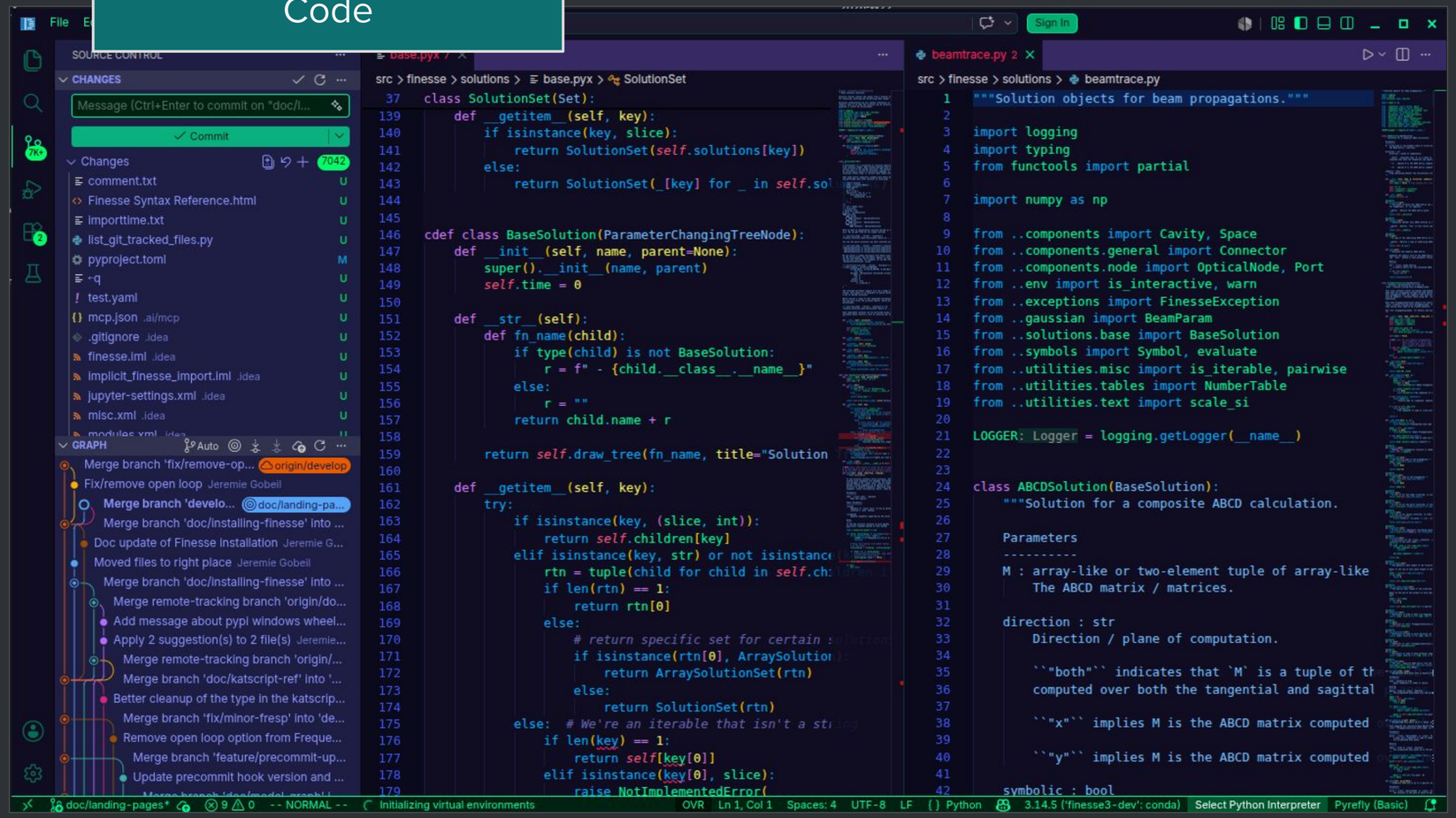
: 3.0rc1! The official Finesse 3

lease. All of the content should be
an't find something. There are links
to previous versions of the documentation on the homepage.

- A very fancy new quick overview syntax page for the KatScript language
- A complete rewrite of the installation instructions

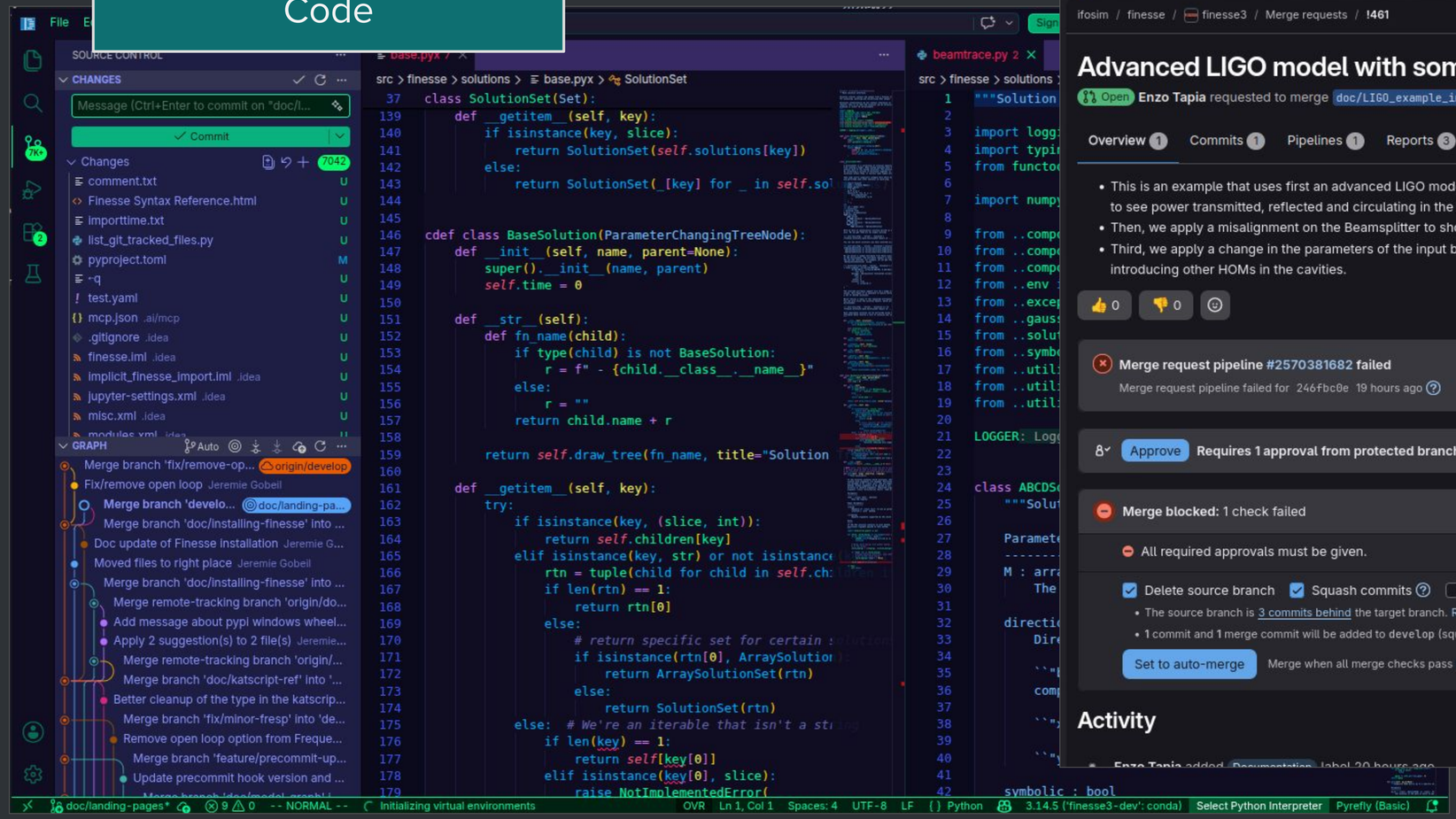
# $ ls --pov=developer ~/finesse3

Code

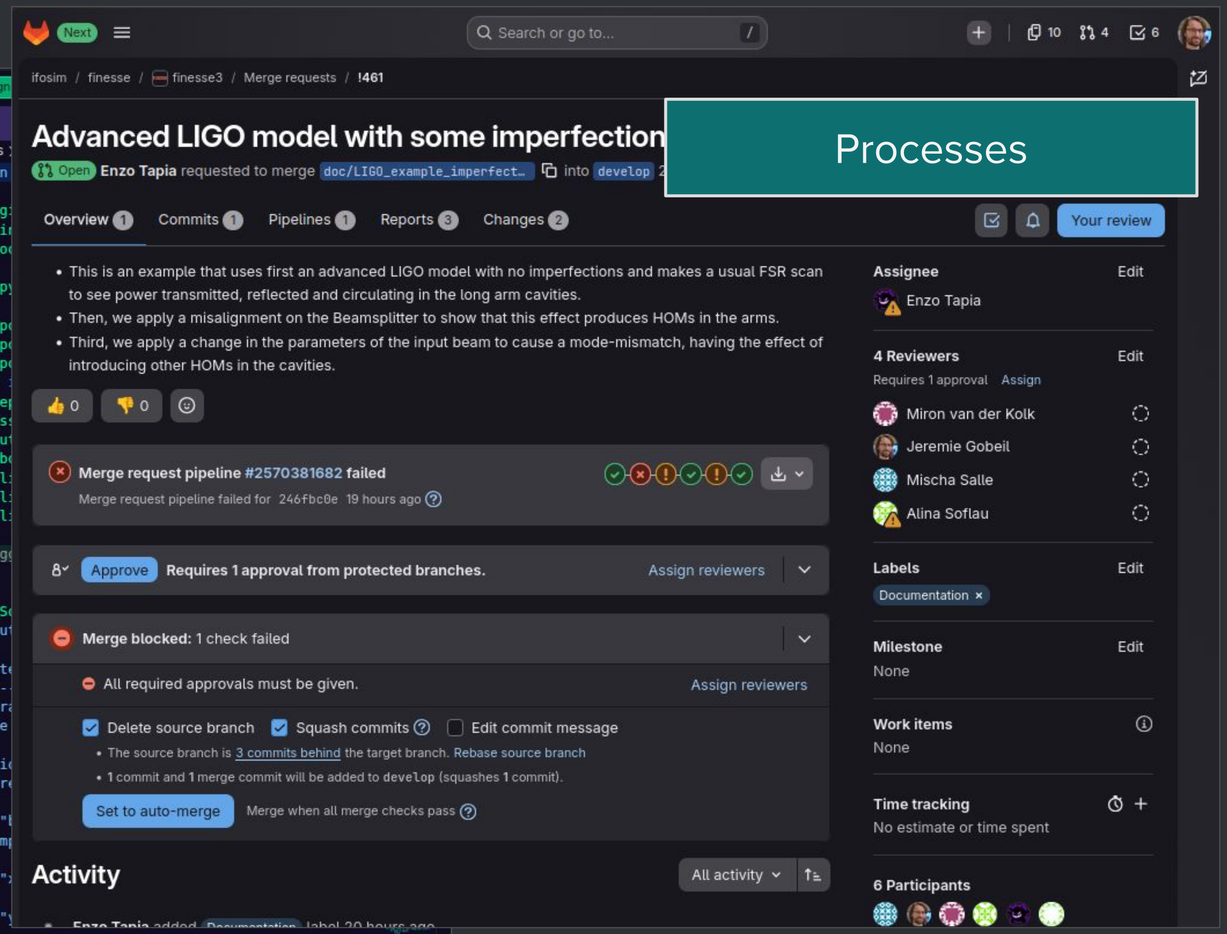


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Code

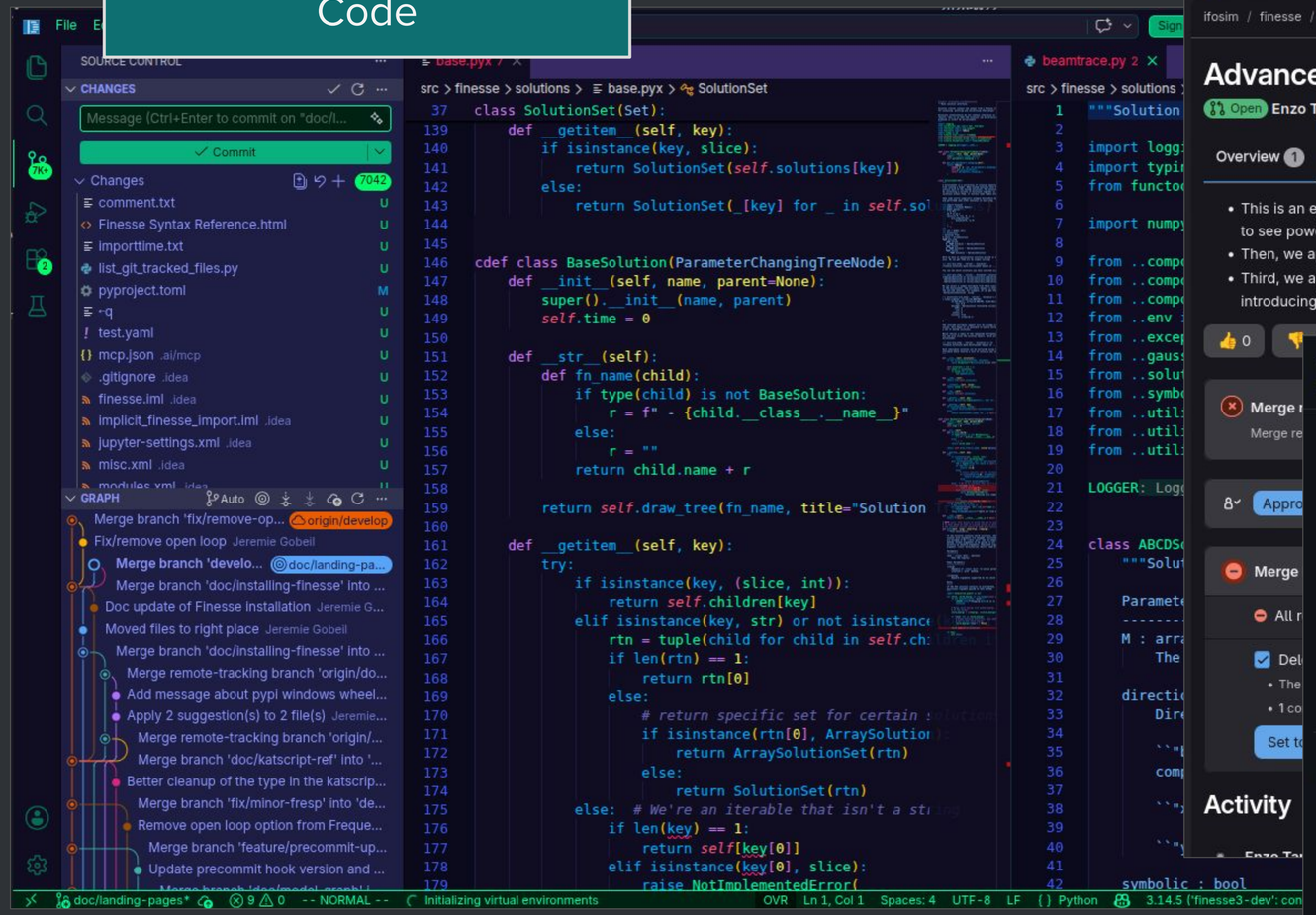



Processes

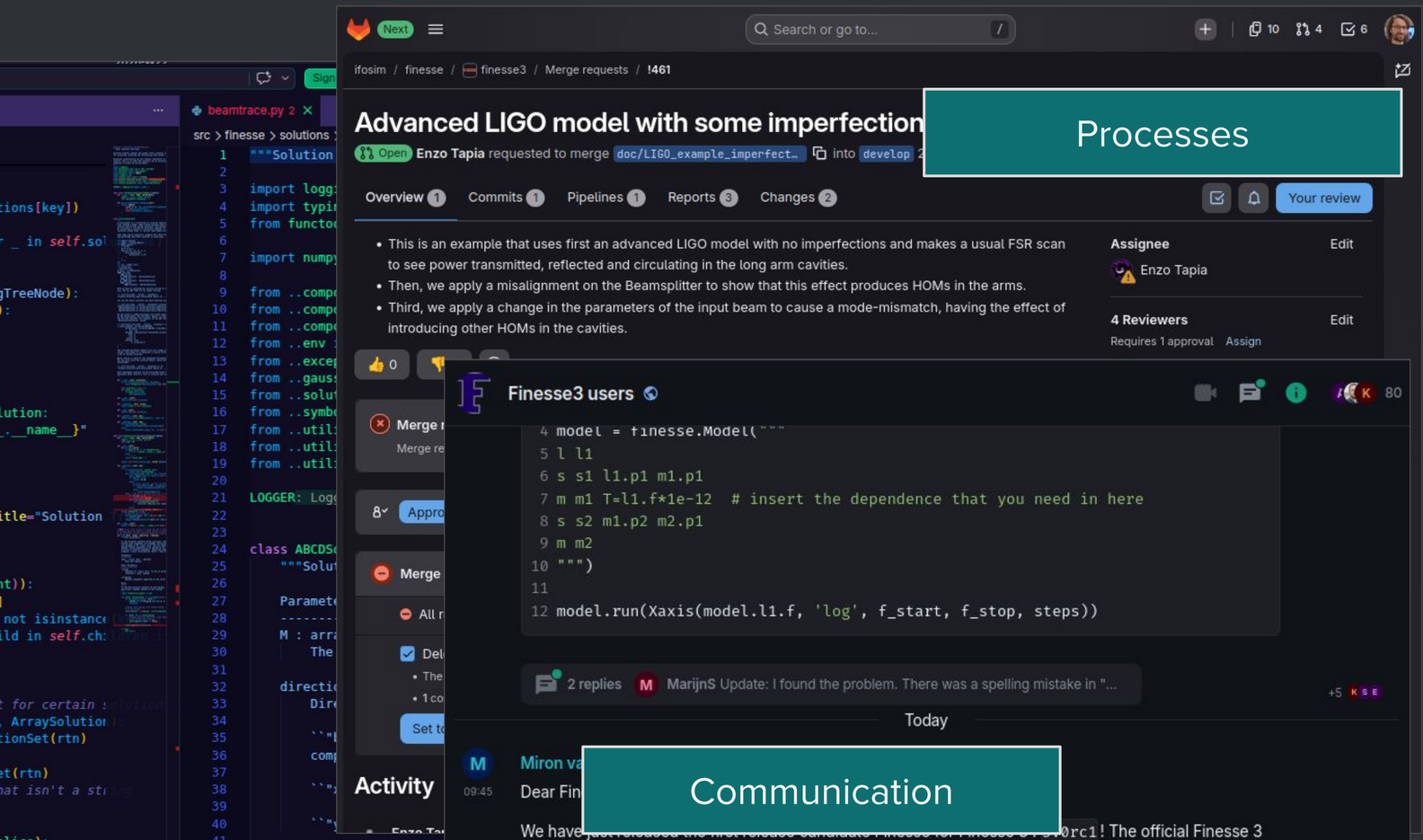


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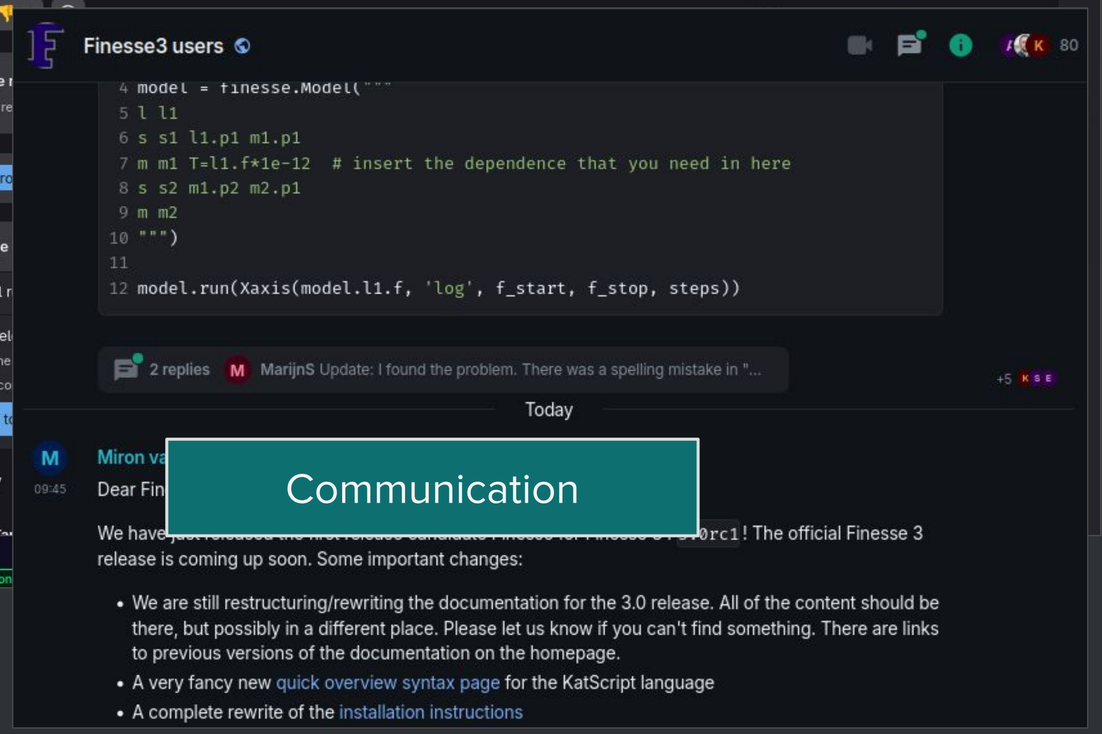
Code



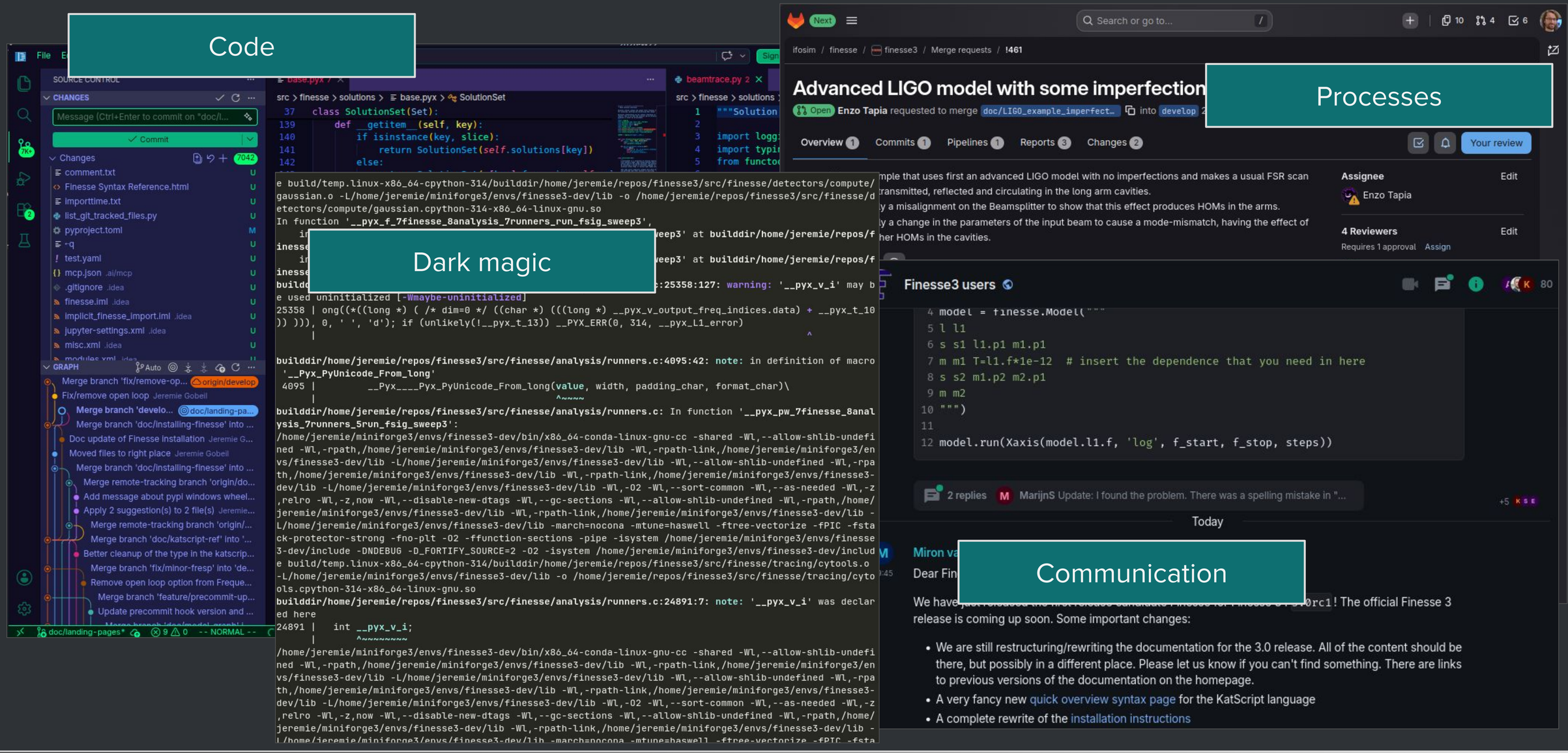

Processes




Communication



# $ ls --pov=developer ~/finesse3



# Code

| | |
|---|---|
| Total Lines of Code | 609940 (3174683 added, 2564743 removed) |
| Total Commits | 6475 (average 5.1 commits per active day, 2.1 per all days) |
| Authors | 29 (average 223.3 commits per author) |

# Code

- Code
- Documentations
- Makefiles
- Test files

| Type | Size | Lines | Commits | Files |
|---|---|---|---|---|
| python | 3.1 MB | 97127 | 5292 | 448 |
| cython | 1.1 MB | 30368 | 2021 | 114 |
| rst | 858.7 KB | 23762 | 916 | 166 |
| yaml | 30.1 KB | 993 | 447 | 6 |
| others | 25.0 MB | 419967 | 332 | 201 |
| ipynb | 2.8 MB | 68986 | 105 | 38 |
| toml | 5.1 KB | 173 | 75 | 1 |
| md | 52.1 KB | 1062 | 45 | 14 |
| git | 2.2 KB | 177 | 39 | 4 |
| css | 42.5 KB | 1542 | 38 | 2 |
| c/c++ | 161.2 KB | 4040 | 22 | 4 |
| js | 5.3 KB | 129 | 3 | 2 |
| sty | 7.0 KB | 194 | 3 | 2 |
| sh | 52 Bytes | 5 | 2 | 1 |
| json | 959 Bytes | 63 | 1 | 1 |

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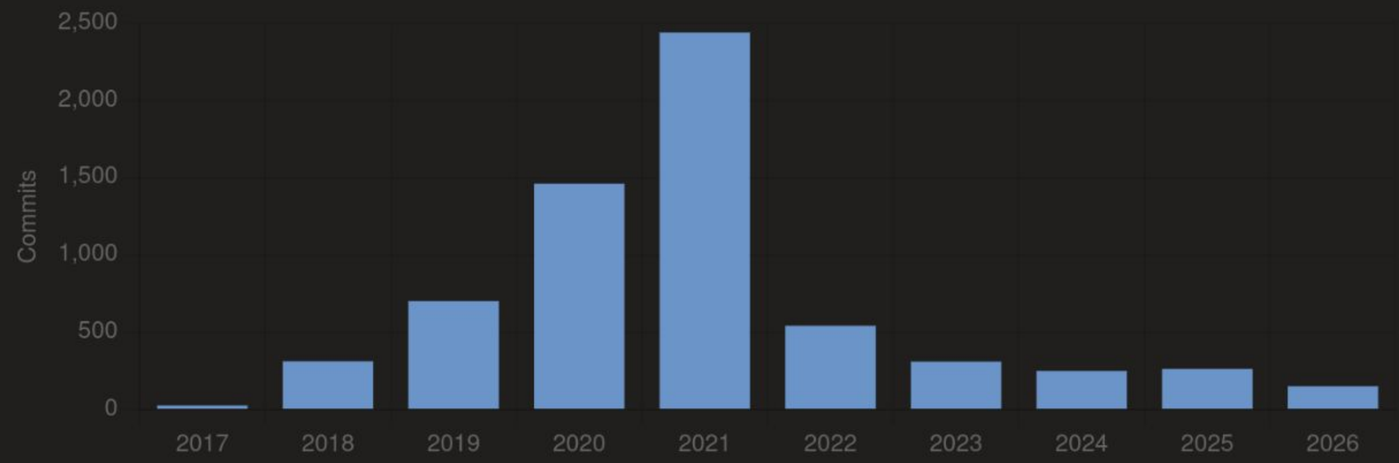
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# Code

## // Commits by Year

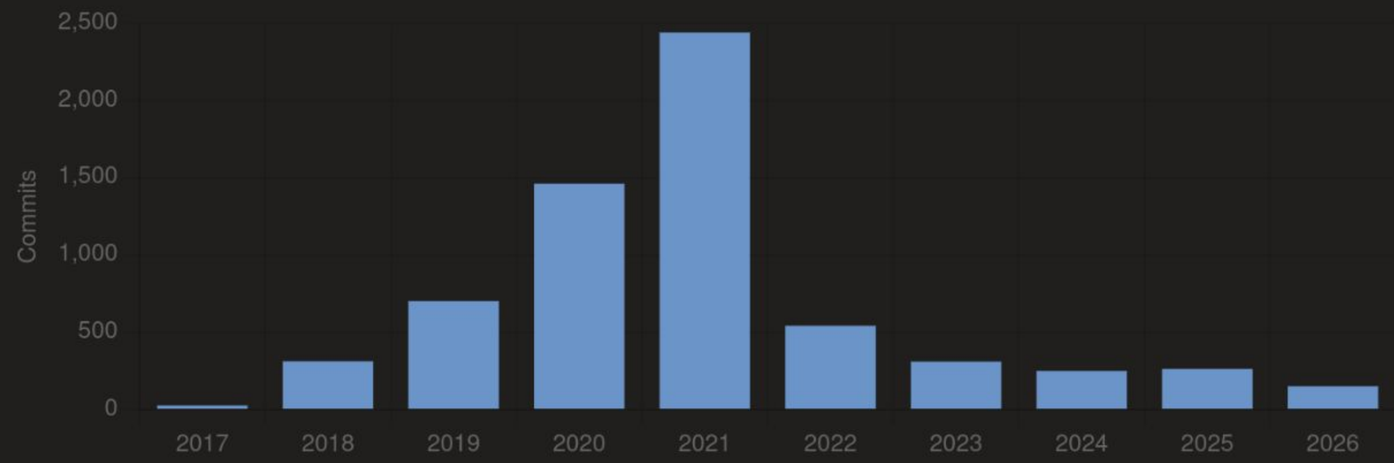
| YEAR | COMMITS (% OF ALL) | LINES ADDED | LINES REMOVED |
|---|---|---|---|
| 2026 | 152 (2.35%) | 86796 | 6881 |
| 2025 | 265 (4.09%) | 14049 | 6522 |
| 2024 | 251 (3.88%) | 27864 | 8862 |
| 2023 | 312 (4.82%) | 12029 | 63144 |
| 2022 | 544 (8.40%) | 25264 | 15732 |
| 2021 | 2442 (37.71%) | 210870 | 128725 |
| 2020 | 1463 (22.59%) | 462362 | 132628 |
| 2019 | 704 (10.87%) | 138637 | 2156757 |
| 2018 | 314 (4.85%) | 2190260 | 41527 |
| 2017 | 28 (0.43%) | 6552 | 3965 |



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Mischa

Miron

Jeremie

# Code

// List of Authors

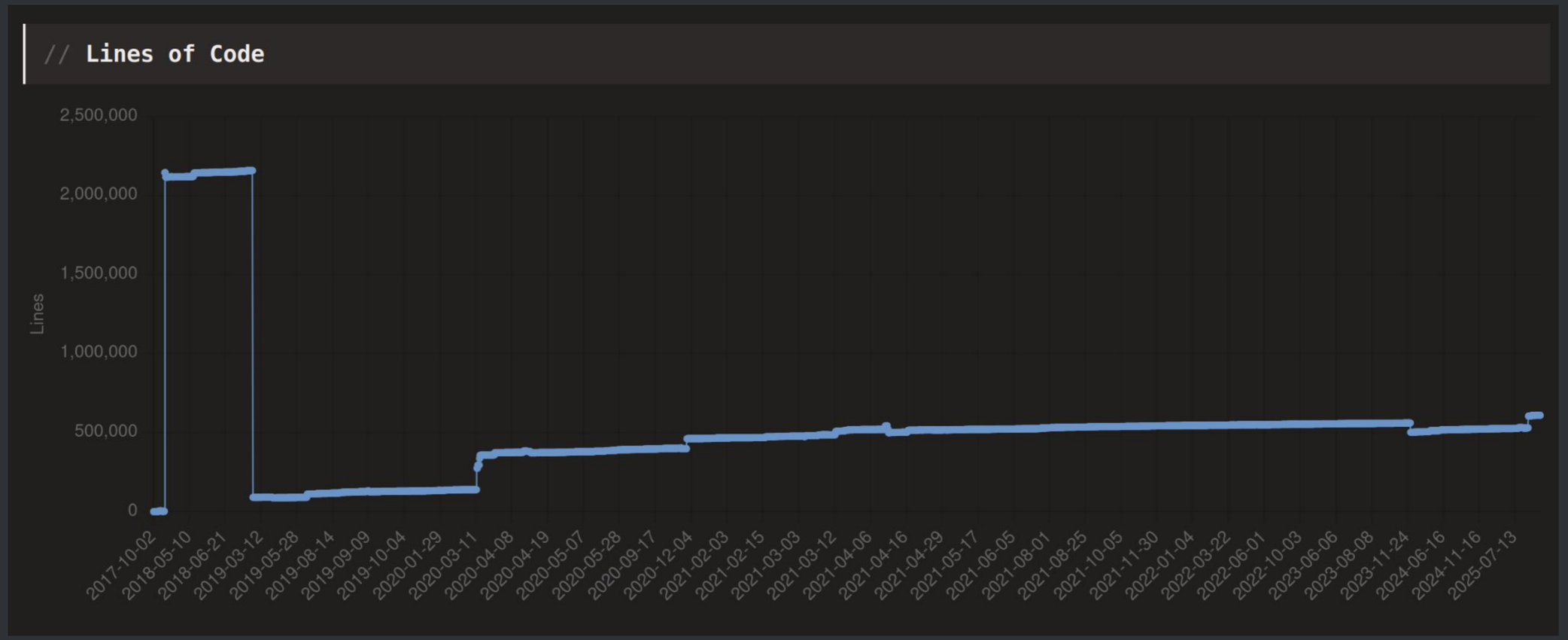
| AUTHOR | COMMITS (%) | + LINES | - LINES | FIRST COMMIT | LAST COMMIT | AGE | ACTIVE DAYS | # BY COMMITS |
|---|---|---|---|---|---|---|---|---|
| Daniel Brown | 2386 (36.85%) | 2299092 | 117196 | 2017-10-05 | 2026-03-16 | 3084 days, 13:26:02 | 598 | 1 |
| Sean Leavey | 1370 (21.16%) | 106510 | 74500 | 2018-03-01 | 2023-09-26 | 2034 days, 20:58:48 | 251 | 2 |
| Samuel Rowlinson | 1052 (16.25%) | 2850114 | 2536223 | 2017-10-02 | 2021-09-28 | 1457 days, 0:03:27 | 487 | 3 |
| Phil Jones | 396 (6.12%) | 58970 | 2172260 | 2018-03-01 | 2021-10-13 | 1322 days, 2:58:29 | 172 | 4 |
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| Jeremie Gobeil | 199 (3.07%) | 273438 | 197844 | 2025-04-15 | 2026-05-22 | 402 days, 0:39:50 | 61 | 6 |
| Andreas Freise | 171 (2.64%) | 49776 | 1653 | 2018-05-14 | 2026-04-22 | 2900 days, 8:50:16 | 63 | 7 |
| Jonathan Perry | 146 (2.25%) | 4689 | 2276 | 2021-07-01 | 2025-03-24 | 1361 days, 17:06:13 | 80 | 8 |
| Aaron Jones | 144 (2.22%) | 93346 | 2527 | 2019-03-07 | 2024-11-13 | 2077 days, 20:19:04 | 51 | 9 |
| Mischa Salle | 102 (1.58%) | 25414 | 1461 | 2021-04-12 | 2026-05-12 | 1855 days, 17:35:04 | 71 | 10 |
| Paul Hapke | 75 (1.16%) | 5465 | 2416 | 2021-10-10 | 2023-11-05 | 756 days, 8:12:30 | 34 | 11 |
| Lee McCuller | 47 (0.73%) | 9961 | 4009 | 2019-08-19 | 2019-09-02 | 14 days, 7:44:58 | 11 | 12 |
| Alexei Ciobanu | 28 (0.43%) | 930 | 433 | 2019-09-12 | 2022-02-03 | 874 days, 22:57:46 | 10 | 13 |
| Sean | 27 (0.42%) | 1134 | 850 | 2019-08-27 | 2020-05-24 | 270 days, 17:25:52 | 4 | 14 |
| Philip Jones | 22 (0.34%) | 6713 | 697 | 2017-10-03 | 2021-04-09 | 1284 days, 0:59:44 | 13 | 15 |
| Kevin Kuns | 21 (0.32%) | 4236 | 1118 | 2023-10-27 | 2026-04-14 | 900 days, 8:46:39 | 19 | 16 |

# Code

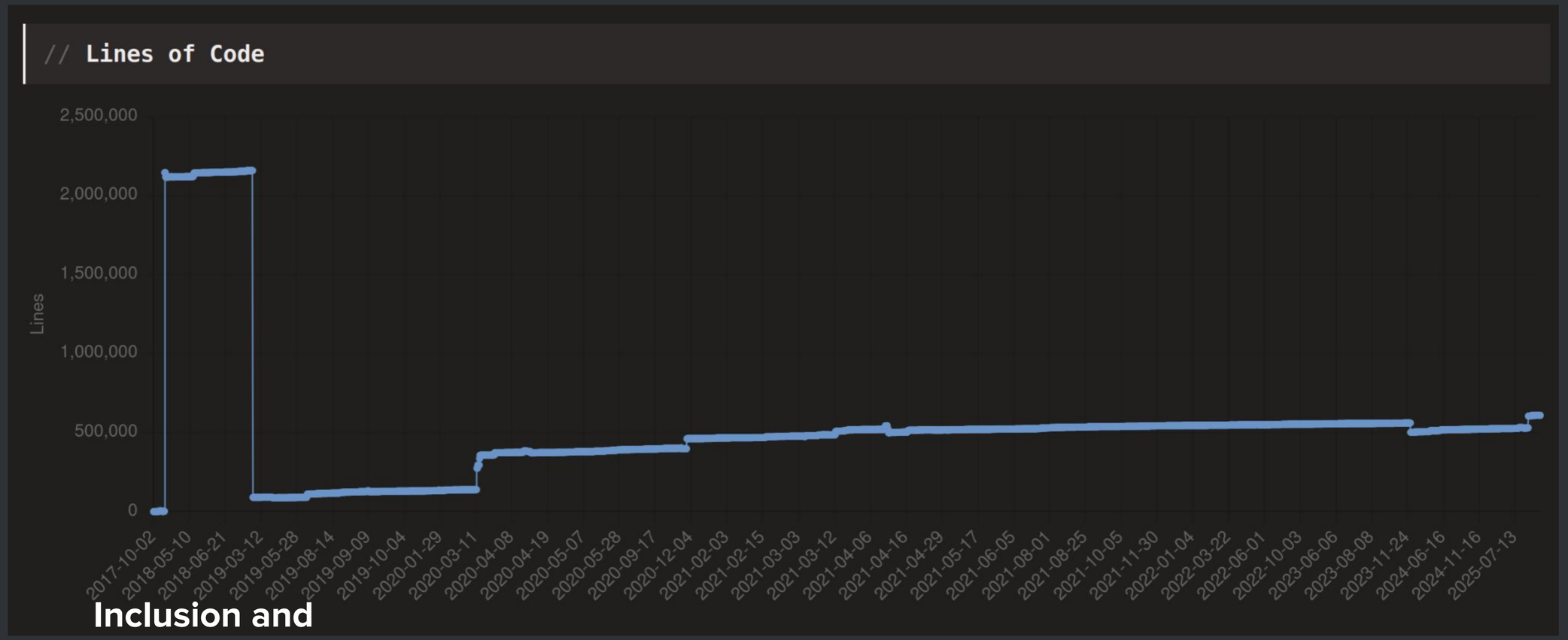
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# Code


// Lines of Code
Lines
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
2017-10-02
2018-05-10
2018-06-21
2019-03-12
2019-05-28
2019-08-14
2019-09-09
2019-10-04
2020-01-29
2020-03-11
2020-04-08
2020-04-19
2020-05-07
2020-05-28
2020-09-17
2020-12-04
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2021-06-05
2021-08-01
2021-08-25
2021-10-05
2021-11-30
2022-01-04
2022-03-22
2022-06-01
2022-10-03
2023-06-06
2023-08-08
2023-11-24
2024-06-16
2024-11-16
2025-07-13

# Code




**Inclusion and removal of SuiteSparse**

# Code

```
// Hour of Week
```

| WEEKDAY | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MON | 11 | 23 | 22 | 23 | 26 | 17 | 22 | 16 | 17 | 65 | 76 | 85 | 69 | 59 | 83 | 103 | 101 | 79 | 51 | 24 | 27 | 30 | 14 | 13 |
| TUE | 20 | 9 | 13 | 35 | 32 | 36 | 17 | 28 | 23 | 76 | 80 | 65 | 62 | 83 | 118 | 95 | 80 | 106 | 74 | 27 | 13 | 22 | 23 | 19 |
| WED | 46 | 28 | 34 | 35 | 13 | 15 | 20 | 22 | 25 | 49 | 64 | 52 | 75 | 80 | 107 | 103 | 113 | 112 | 77 | 49 | 33 | 30 | 34 | 23 |
| THU | 14 | 25 | 49 | 18 | 26 | 34 | 17 | 22 | 26 | 90 | 82 | 44 | 54 | 90 | 120 | 121 | 96 | 98 | 77 | 31 | 40 | 39 | 29 | 34 |
| FRI | 15 | 19 | 13 | 21 | 20 | 25 | 23 | 15 | 30 | 46 | 78 | 75 | 82 | 81 | 55 | 54 | 74 | 85 | 52 | 12 | 16 | 22 | 24 | 20 |
| SAT | 4 | 6 | 8 | 11 | 19 | 25 | 12 | 14 | 23 | 19 | 12 | 14 | 18 | 27 | 25 | 24 | 18 | 4 | 9 | 8 | 8 | 4 | 7 | 8 |
| SUN | 8 | 6 | 13 | 10 | 16 | 18 | 7 | 22 | 11 | 19 | 36 | 39 | 30 | 39 | 26 | 24 | 30 | 21 | 15 | 19 | 16 | 19 | 11 | 9 |

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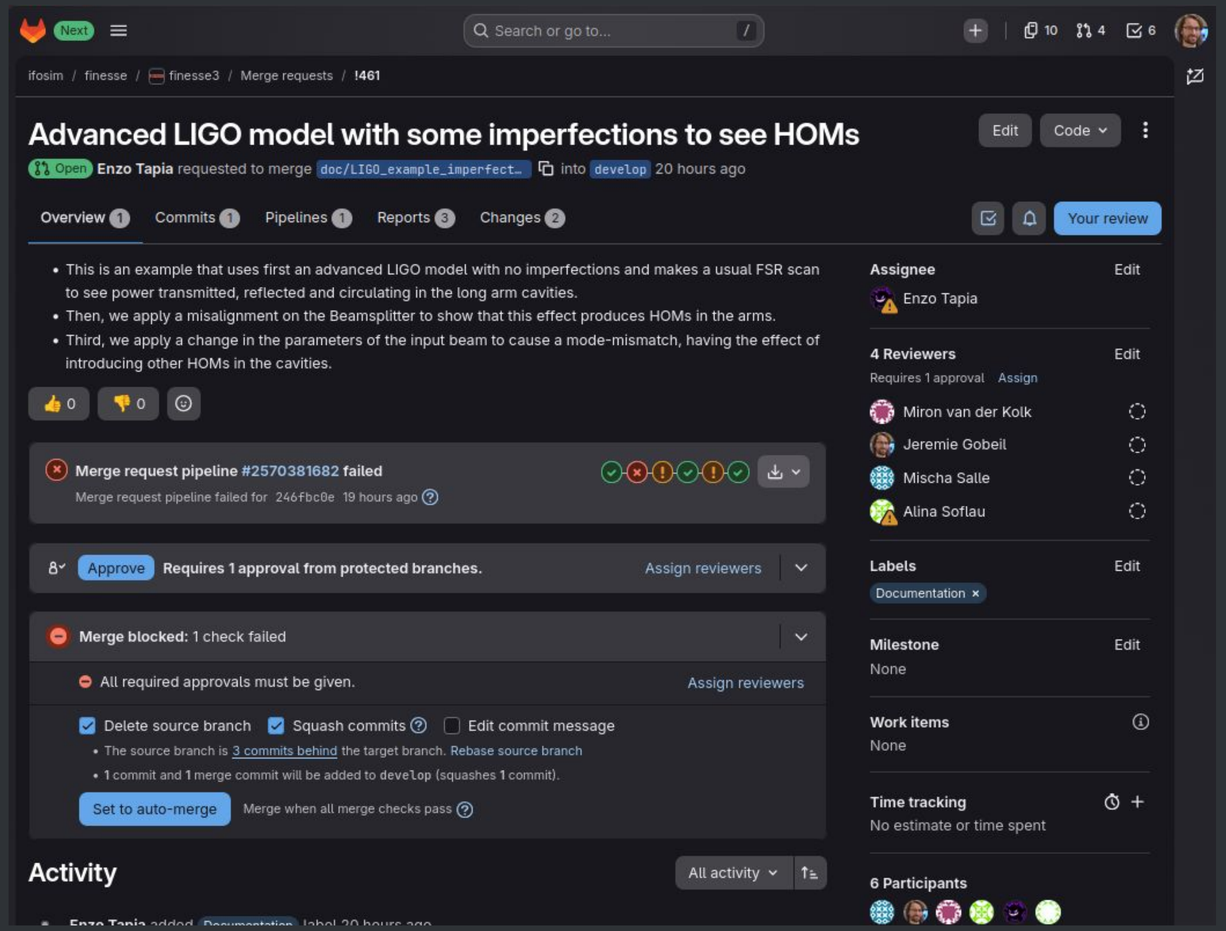
## // Hour of Week

| WEEKDAY | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MON | 11 | 23 | 22 | 23 | 26 | 17 | 22 | 16 | 17 | 65 | 76 | 85 | 69 | 59 | 83 | 103 | 101 | 79 | 51 | 24 | 27 | 30 | 14 | 13 |
| TUE | 20 | 9 | 13 | 35 | 32 | 36 | 17 | 28 | 23 | 76 | 80 | 65 | 62 | 83 | 118 | 95 | 80 | 106 | 74 | 27 | 13 | 22 | 23 | 19 |
| WED | 46 | 28 | 34 | 35 | 13 | 15 | 20 | 22 | 25 | 49 | 64 | 52 | 75 | 80 | 107 | 103 | 113 | 112 | 77 | 49 | 33 | 30 | 34 | 23 |
| THU | 14 | 25 | 49 | 18 | 26 | 34 | 17 | 22 | 26 | 90 | 82 | 44 | 54 | 90 | 120 | 121 | 96 | 98 | 77 | 31 | 40 | 39 | 29 | 34 |
| FRI | 15 | 19 | 13 | 21 | 20 | 25 | 23 | 15 | 30 | 46 | 78 | 75 | 82 | 81 | 55 | 54 | 74 | 85 | 52 | 12 | 16 | 22 | 24 | 20 |
| SAT | 4 | 6 | 8 | 11 | 19 | 25 | 12 | 14 | 23 | 19 | 12 | 14 | 18 | 27 | 25 | 24 | 18 | 4 | 9 | 8 | 8 | 4 | 7 | 8 |
| SUN | | | | | | | | | | | | | | | | | | | | | | 19 | 11 | 9 |

## // Commits by Timezone

| -0800 | -0700 | -0600 | -0500 | -0400 | +0000 | +0100 | +0200 | +0300 | +0400 | +0800 | +0930 | +1000 | +1030 | +1100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 63 | 59 | 25 | 44 | 49 | 1225 | 1600 | 1305 | 4 | 1 | 5 | 1222 | 13 | 846 | 14 |

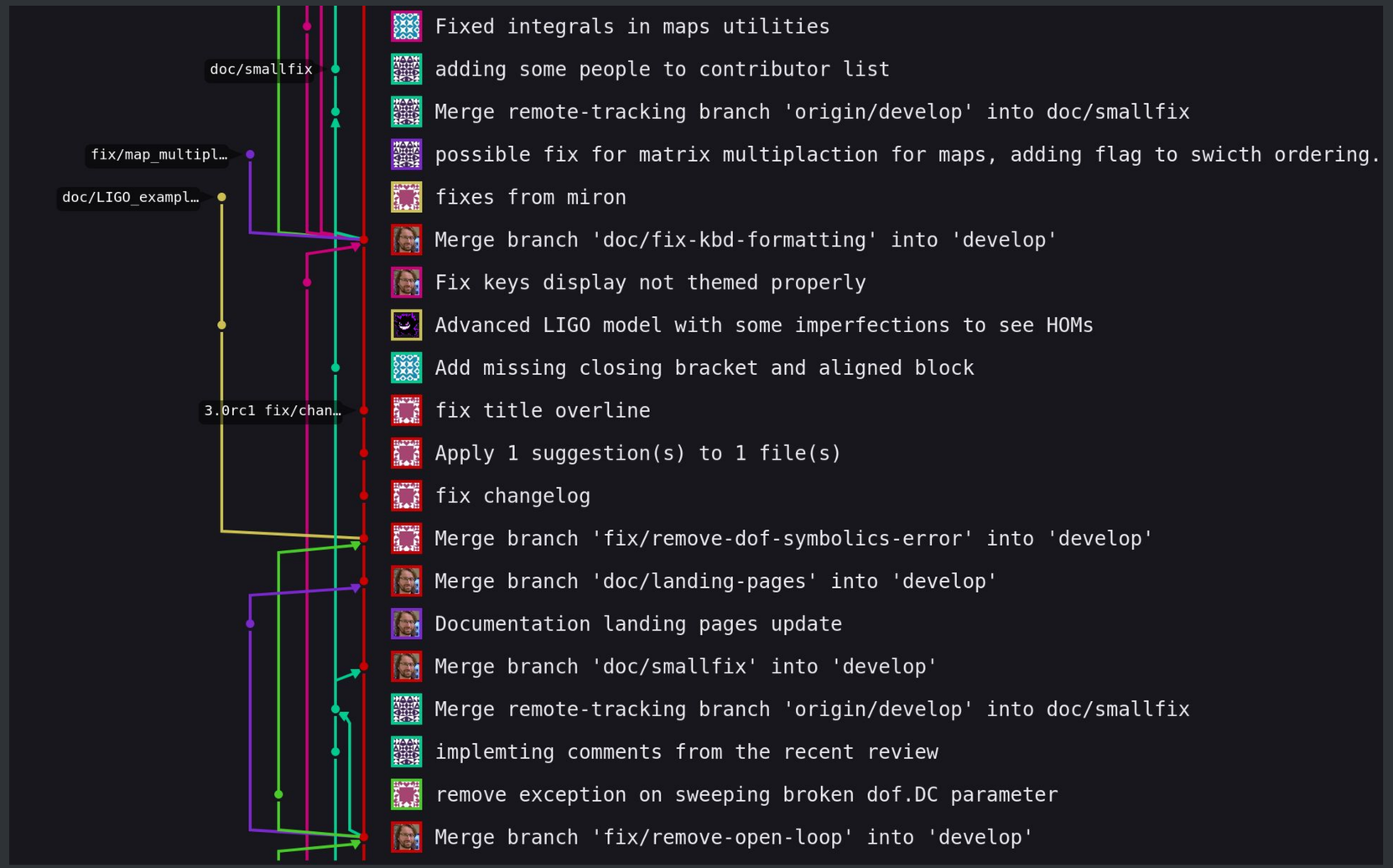
# Processes




- Git repository
- GitLab
- Issues
- Merge requests
- Code reviews
- CI/CD (pipelines)

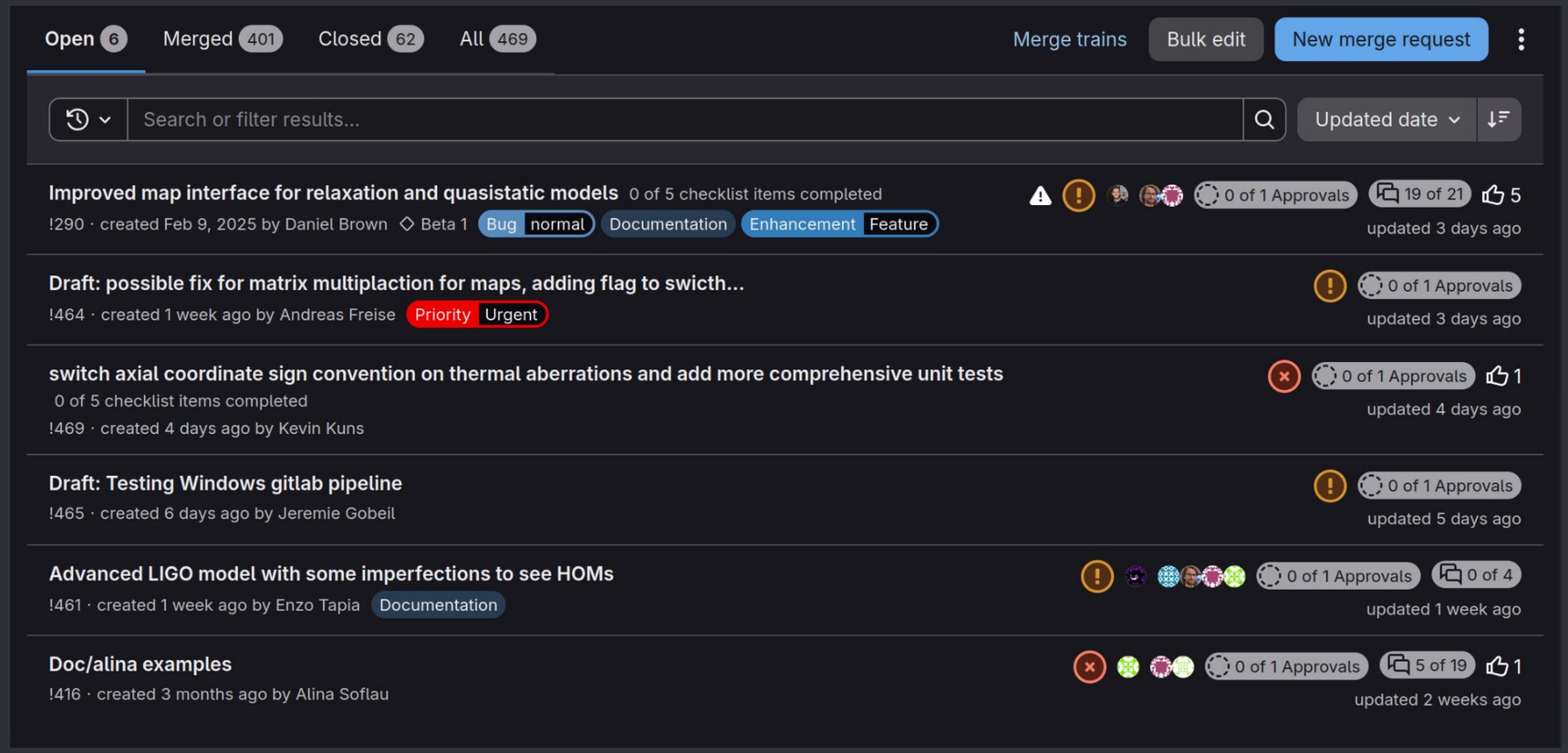
Open source best practice (adapted)

# Processes



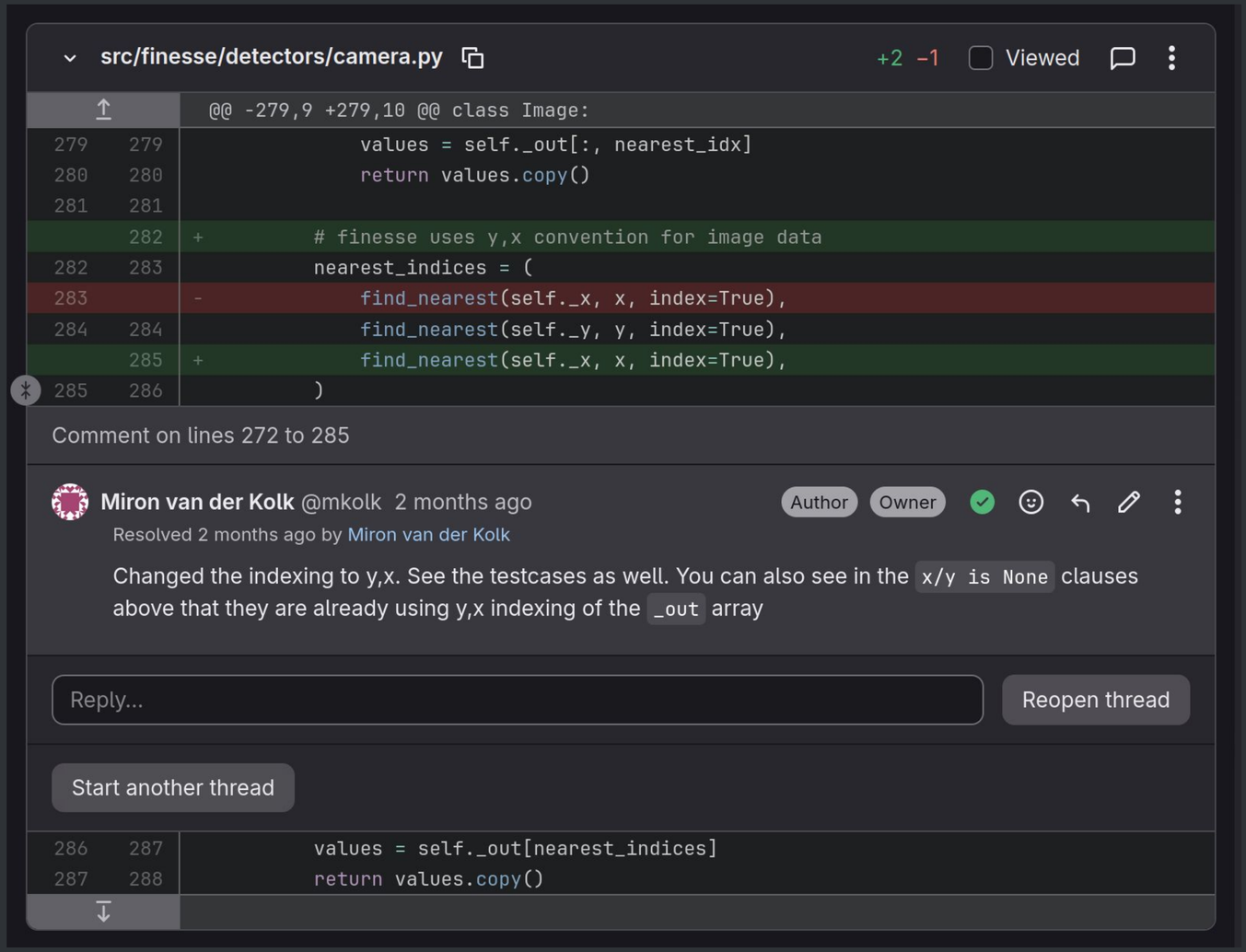

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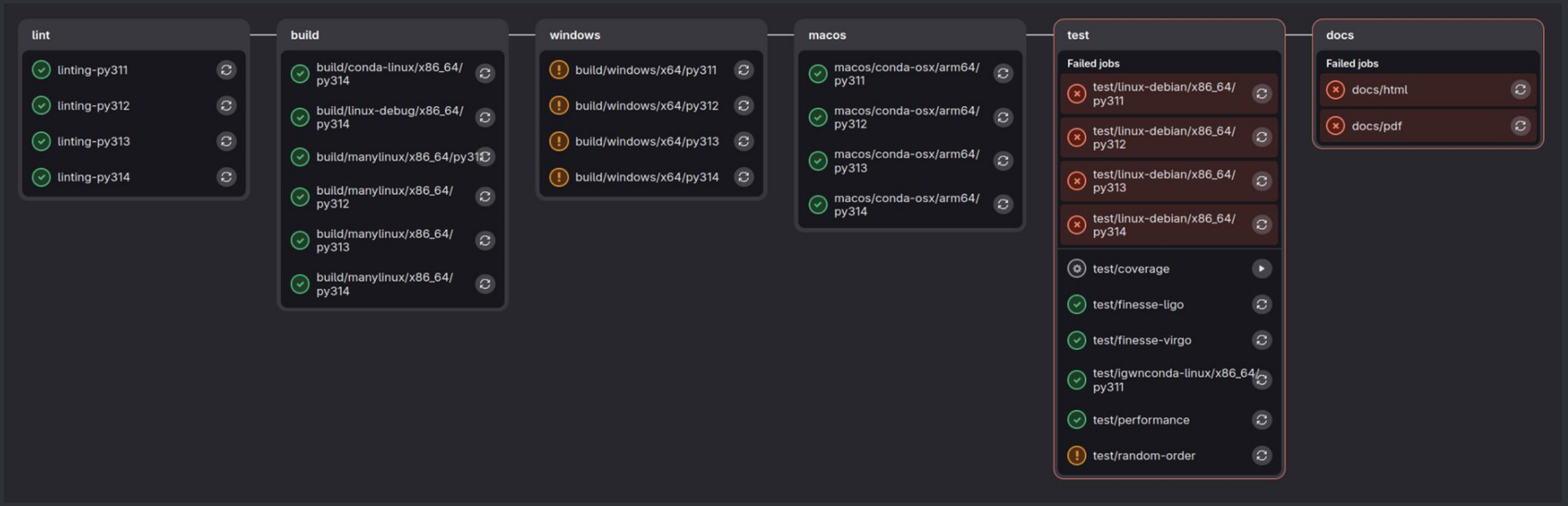

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- CI/CD (pipelines)

# Processes



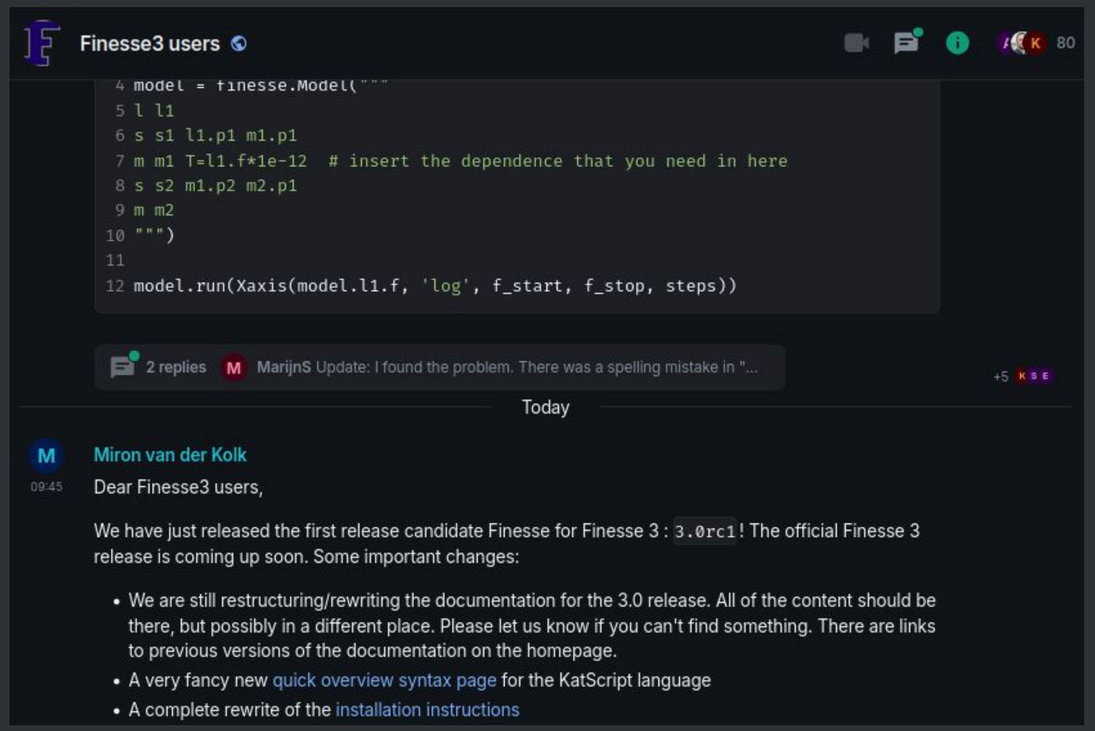

- Git repository
- GitLab
- Issues
- Merge requests
- **Code reviews**
- CI/CD (pipelines)

# Processes




- Git repository
- GitLab
- Issues
- Merge requests
- Code reviews
- **CI/CD (pipelines)**

# Communication




- Communication channels
- Meetings
- Workshops
- Tutorials

https://matrix.to/#/#finesse:matrix.org [Matrix]

# Dark magic

```
e build/temp.linux-x86_64-cpython-314/builddir/home/jeremie/repos/finesse3/src/finesse/detectors/compute/
gaussian.o -L/home/jeremie/miniforge3/envs/finesse3-dev/lib -o /home/jeremie/repos/finesse3/src/finesse/d
etectors/compute/gaussian.cpython-314-x86_64-linux-gnu.so
In function '__pyx_f_7finesse_8analysis_7runners_run_fsig_sweep3',
    inlined from '__pyx_pf_7finesse_8analysis_7runners_4run_fsig_sweep3' at builddir/home/jeremie/repos/f
inesse3/src/finesse/analysis/runners.c:26402:15,
    inlined from '__pyx_pw_7finesse_8analysis_7runners_5run_fsig_sweep3' at builddir/home/jeremie/repos/f
inesse3/src/finesse/analysis/runners.c:26354:13:
builddir/home/jeremie/repos/finesse3/src/finesse/analysis/runners.c:25358:127: warning: '__pyx_v_i' may b
e used uninitialized [-Wmaybe-uninitialized]
25358 | ong((*((long *) ( /* dim=0 */ ((char *) (((long *) __pyx_v_output_freq_indices.data) + __pyx_t_10
)) ))), 0, ' ', 'd'); if (unlikely(!__pyx_t_13)) __PYX_ERR(0, 314, __pyx_L1_error)
      |                                                                                                 ^

builddir/home/jeremie/repos/finesse3/src/finesse/analysis/runners.c:4095:42: note: in definition of macro
 '__Pyx_PyUnicode_From_long'
 4095 |         __Pyx____Pyx_PyUnicode_From_long(value, width, padding_char, format_char)\
      |                                          ^~~~~
builddir/home/jeremie/repos/finesse3/src/finesse/analysis/runners.c: In function '__pyx_pw_7finesse_8anal
ysis_7runners_5run_fsig_sweep3':
/home/jeremie/miniforge3/envs/finesse3-dev/bin/x86_64-conda-linux-gnu-cc -shared -Wl,--allow-shlib-undefi
ned -Wl,-rpath,/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/en
vs/finesse3-dev/lib -L/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,--allow-shlib-undefined -Wl,-rpa
th,/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/envs/finesse3-
dev/lib -L/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-O2 -Wl,--sort-common -Wl,--as-needed -Wl,-z
,relro -Wl,-z,now -Wl,--disable-new-dtags -Wl,--gc-sections -Wl,--allow-shlib-undefined -Wl,-rpath,/home/
jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/envs/finesse3-dev/lib -
L/home/jeremie/miniforge3/envs/finesse3-dev/lib -march=nocona -mtune=haswell -ftree-vectorize -fPIC -fsta
ck-protector-strong -fno-plt -O2 -ffunction-sections -pipe -isystem /home/jeremie/miniforge3/envs/finesse
3-dev/include -DNDEBUG -D_FORTIFY_SOURCE=2 -O2 -isystem /home/jeremie/miniforge3/envs/finesse3-dev/includ
e build/temp.linux-x86_64-cpython-314/builddir/home/jeremie/repos/finesse3/src/finesse/tracing/cytools.o
-L/home/jeremie/miniforge3/envs/finesse3-dev/lib -o /home/jeremie/repos/finesse3/src/finesse/tracing/cyto
ols.cpython-314-x86_64-linux-gnu.so
builddir/home/jeremie/repos/finesse3/src/finesse/analysis/runners.c:24891:7: note: '__pyx_v_i' was declar
ed here
24891 |   int __pyx_v_i;
      |       ^~~~~~~~~
/home/jeremie/miniforge3/envs/finesse3-dev/bin/x86_64-conda-linux-gnu-cc -shared -Wl,--allow-shlib-undefi
ned -Wl,-rpath,/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/en
vs/finesse3-dev/lib -L/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,--allow-shlib-undefined -Wl,-rpa
th,/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/envs/finesse3-
dev/lib -L/home/jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-O2 -Wl,--sort-common -Wl,--as-needed -Wl,-z
,relro -Wl,-z,now -Wl,--disable-new-dtags -Wl,--gc-sections -Wl,--allow-shlib-undefined -Wl,-rpath,/home/
jeremie/miniforge3/envs/finesse3-dev/lib -Wl,-rpath-link,/home/jeremie/miniforge3/envs/finesse3-dev/lib -
L/home/jeremie/miniforge3/envs/finesse3-dev/lib -march=nocona -mtune=haswell -ftree-vectorize -fPIC -fsta
```

- Servers to maintain
- Dependencies to manage (conflicts, errors, ...)
- Package distribution channels to
- Compilation on Windows (why?)

Player 2
Press key to start...

# Finesse : how does it work?

# A FINESSE simulation



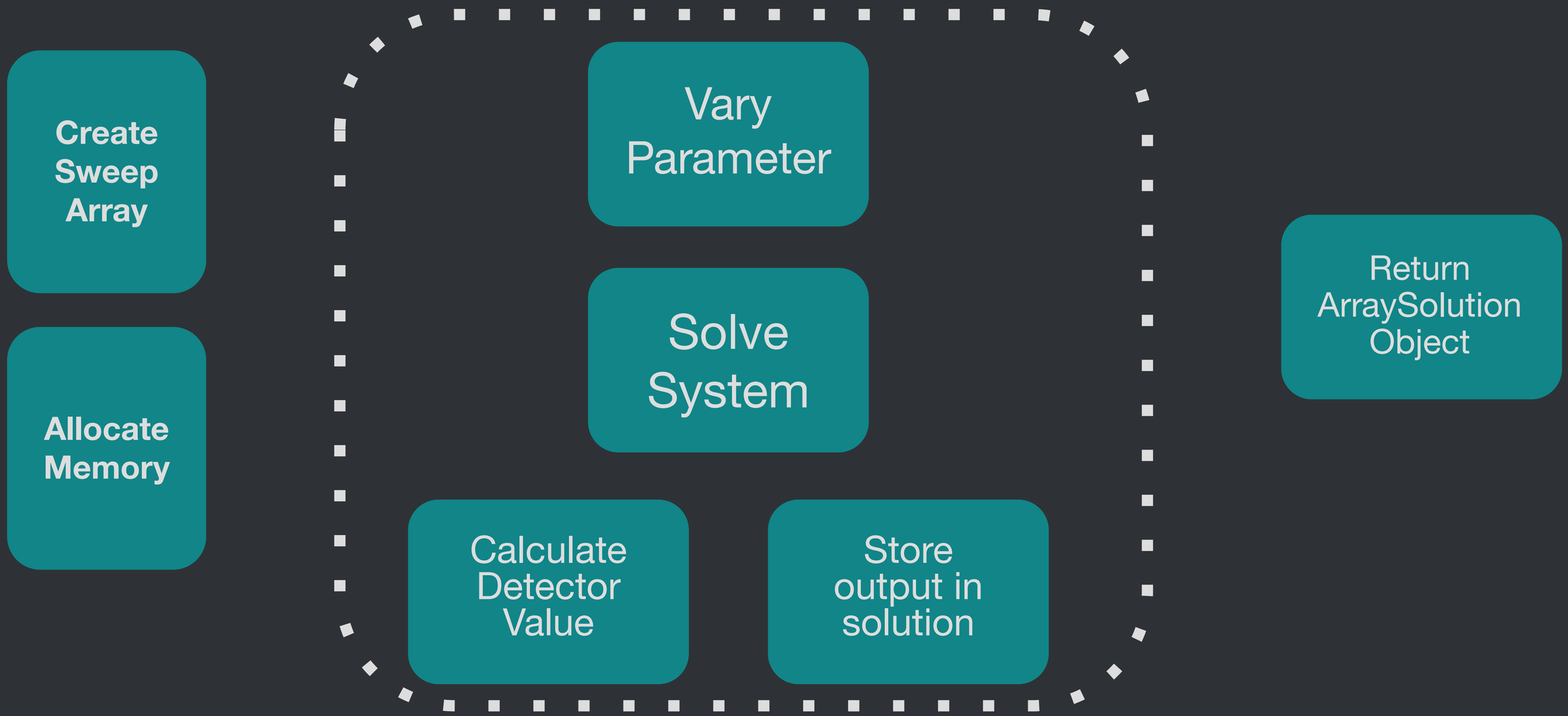
# Building the model




- Parsing the KatScript
- Which component exist
- What are their physical parameters
- How are they connected
- Adjust model settings (which physics are included)
- Build system of linear equations (matrix)

# Running an action: Xaxis

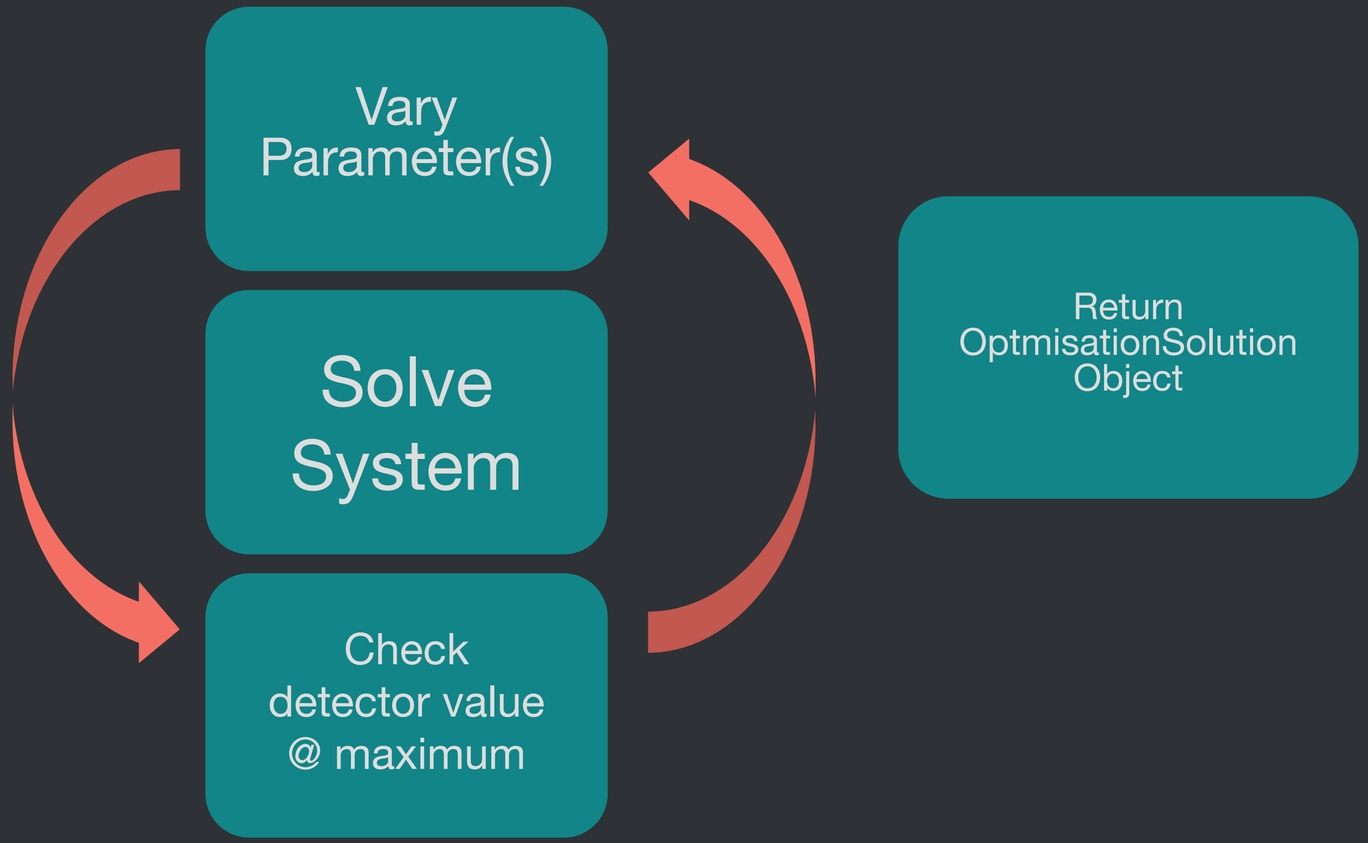


# Running an action: Maximize




- Decide which parameters to vary
- Decide which detector value to maximize



# Returning a Solution



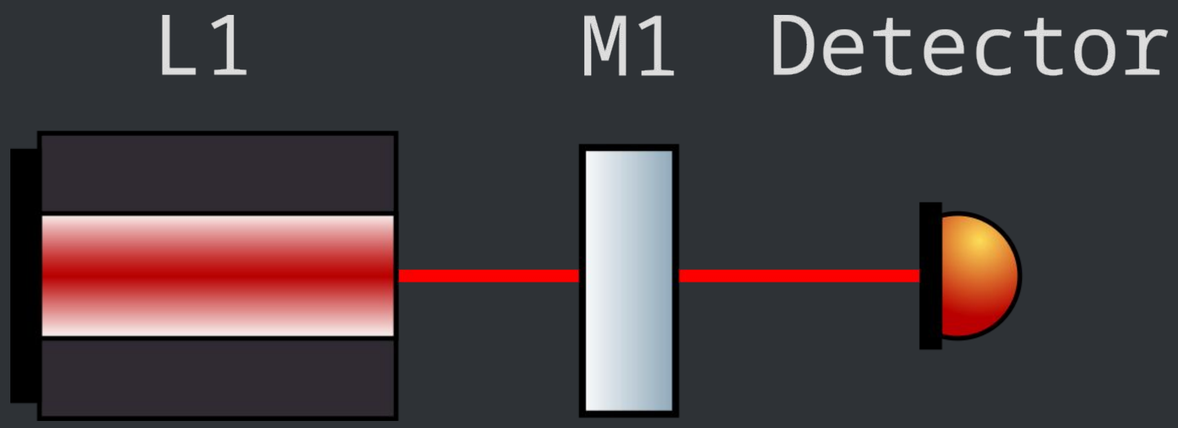

- Xaxis ⇒ ArraySolution: Contains arrays for all detectors, plotting functions
- Maximize ⇒ Final parameter values and optimization log

# Some examples in detail

- **Plane wave**
- Modulation frequencies
- Higher-order modes
- Transfer function / signal modelling

# Plane Wave Setup

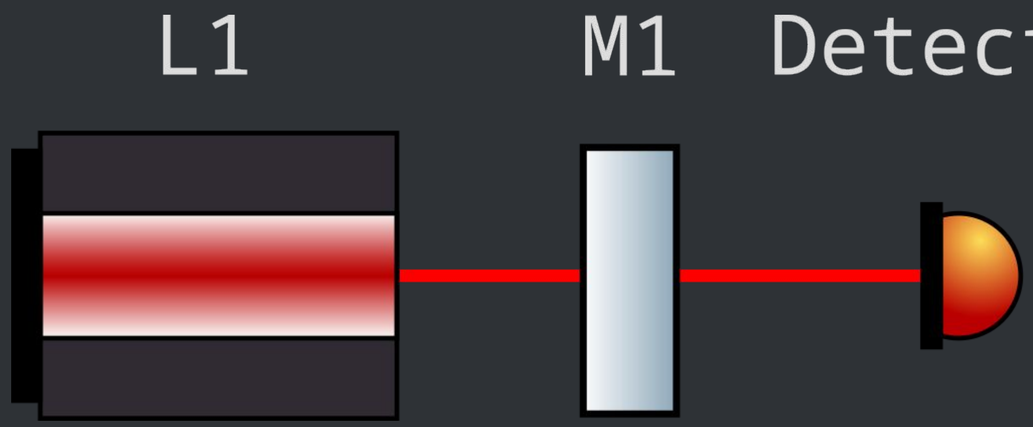
1. Sketching the model

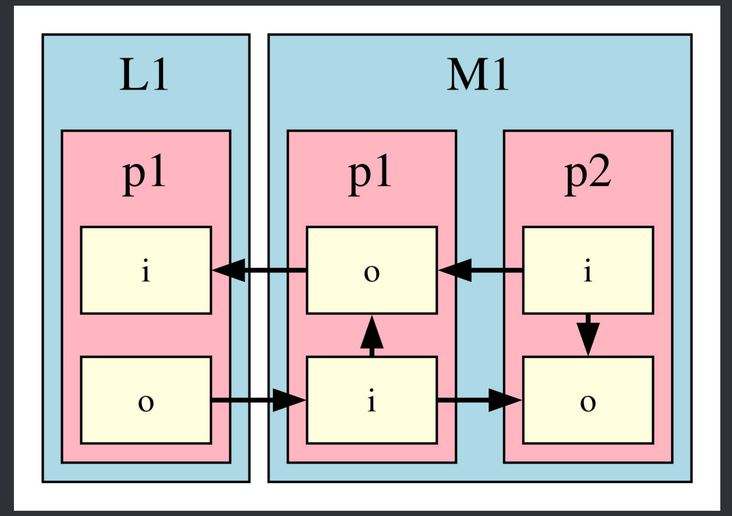



2. Defining in Katscript

```
laser L1 P=1
space S1 L1.p1 M1.p1
mirror M1 R=0.9 T=0.1
pd transmitted_power M1.p2.o
```

# 3. Building the “interferometer” Matrix

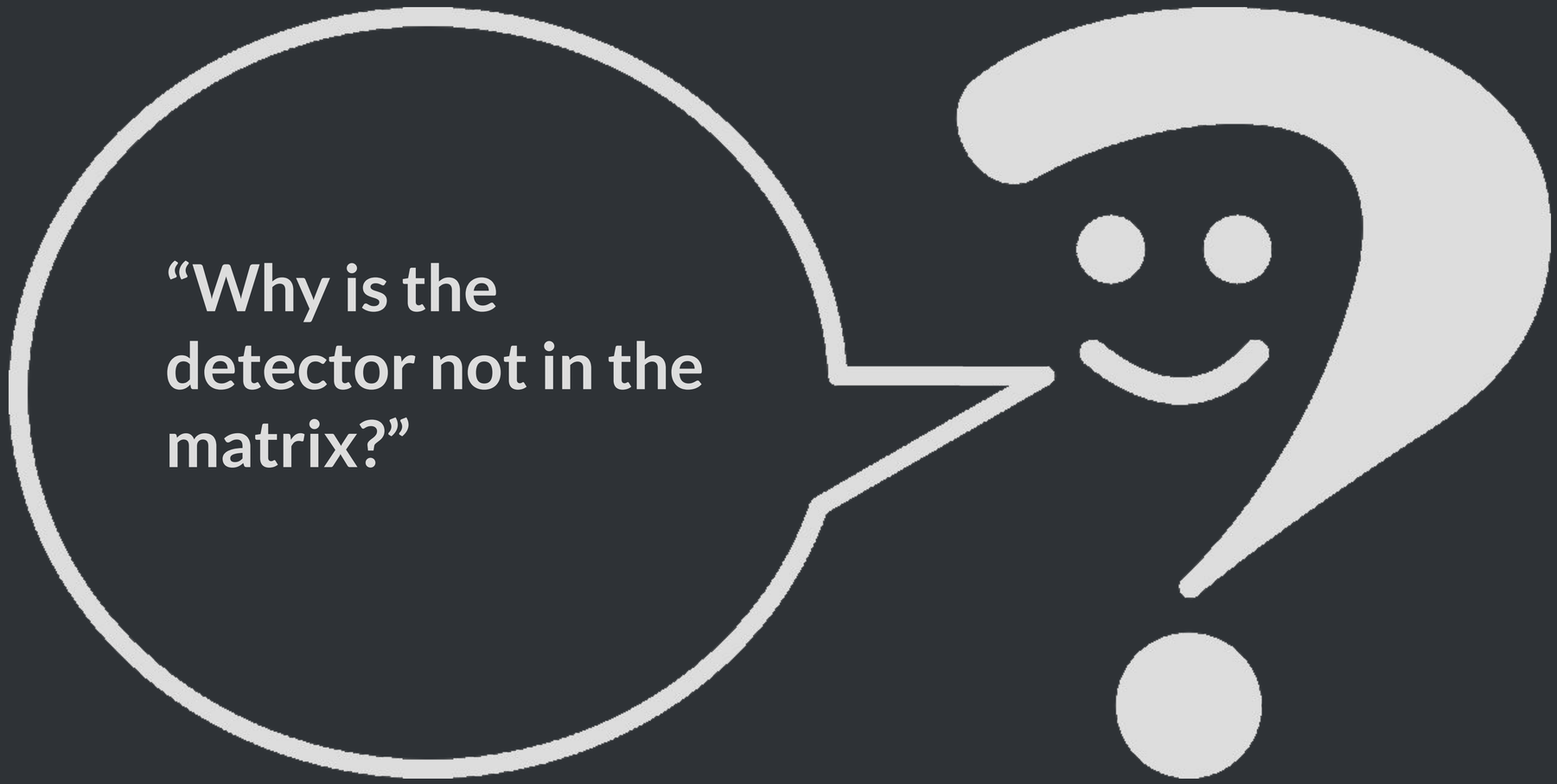



**Interferometer Matrix** **Solution** **Input/RHS**

$$\begin{pmatrix} 1 & 0 & 0 & \rightarrow & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \rightarrow & 1 & 0 & 0 & 0 \\ 0 & 0 & \circlearrowright & 1 & \leftarrow & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \rightarrow & 0 & \circlearrowright & 1 \end{pmatrix} \times \begin{pmatrix} \leftarrow \text{L1.p1.i} \\ \rightarrow \text{L1.p1.o} \\ \rightarrow \text{M1.p1.i} \\ \leftarrow \text{M1.p1.o} \\ \leftarrow \text{M1.p2.i} \\ \rightarrow \text{M1.p2.o} \end{pmatrix} = \begin{pmatrix} 0 \\ 1.0+0j \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

# Question from Jeremie

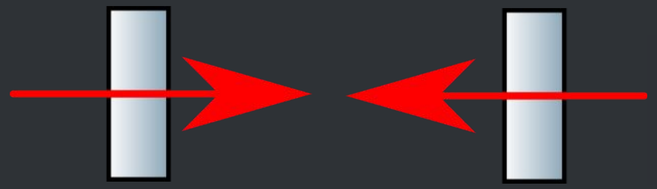


# 4. Filling the matrix

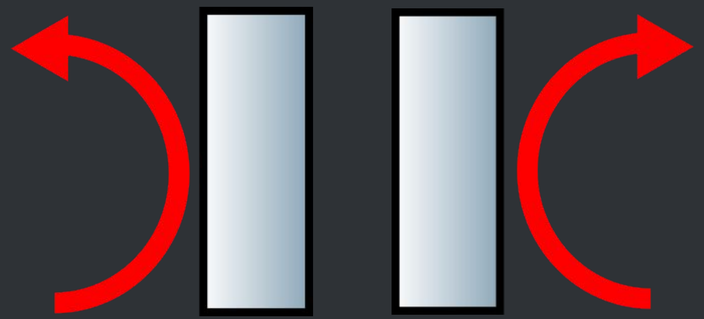


Space propagation:
Macroscopic length

$$\exp\left(-ikd\right)$$

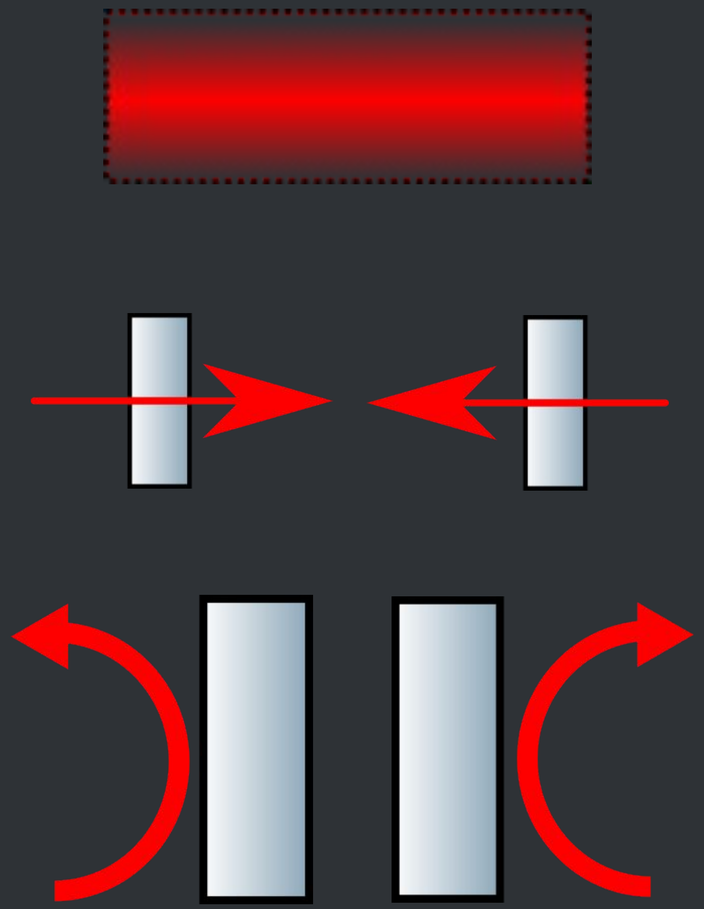


$$M_{12} = M_{21} = te^{i\varphi_{12}(\phi)} = te^{i\varphi_{21}(\phi)}$$



$$M_{11} = re^{i\varphi_{11}(\phi)}$$
$$M_{22} = re^{i\varphi_{22}(\phi)}$$

# 4. Filling the matrix



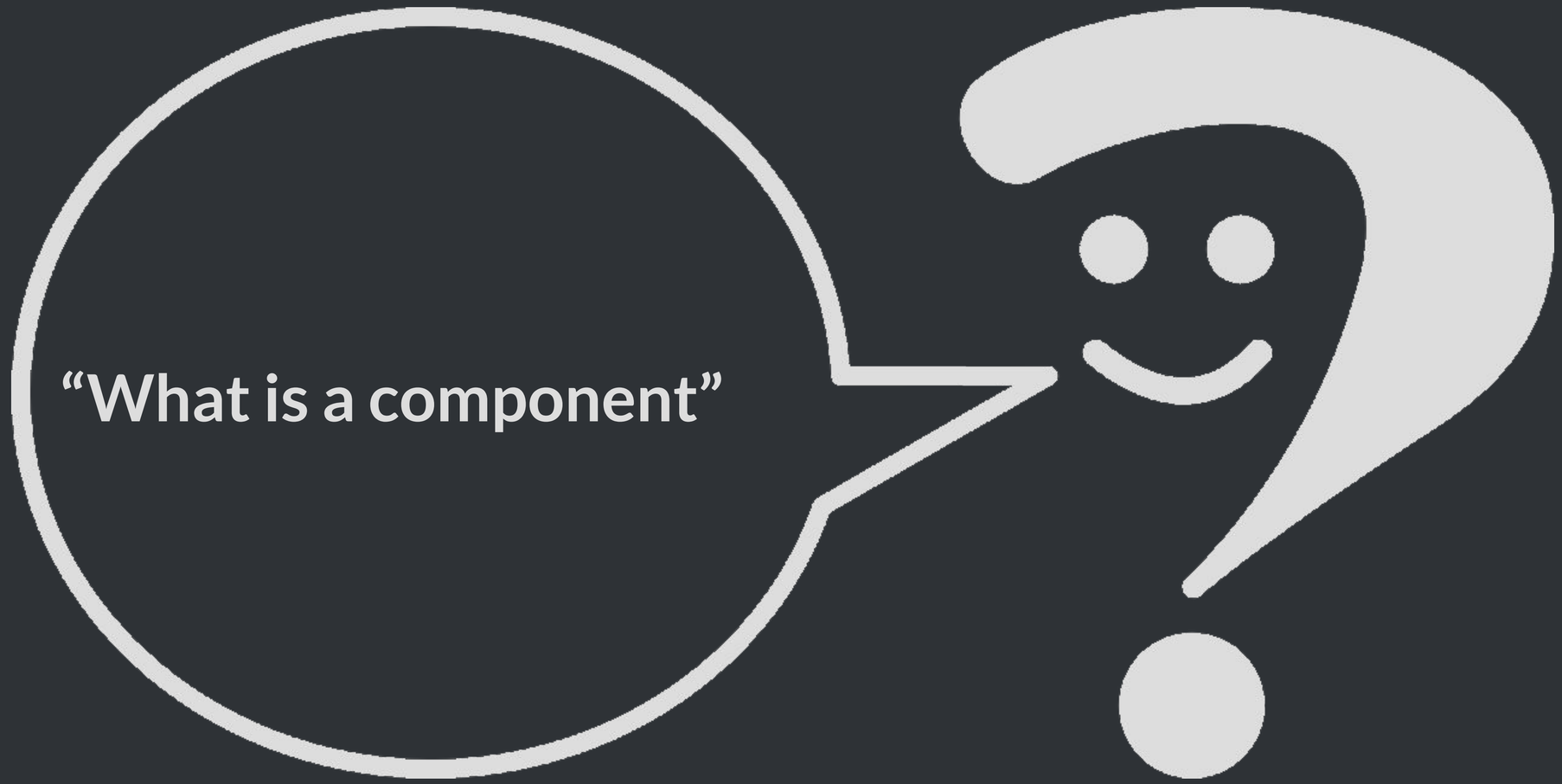
Calculate the linear couplings for:

- Space Propagation
- Mirror Transmission
- Mirror Reflection

Equations depend on component *Parameters:*

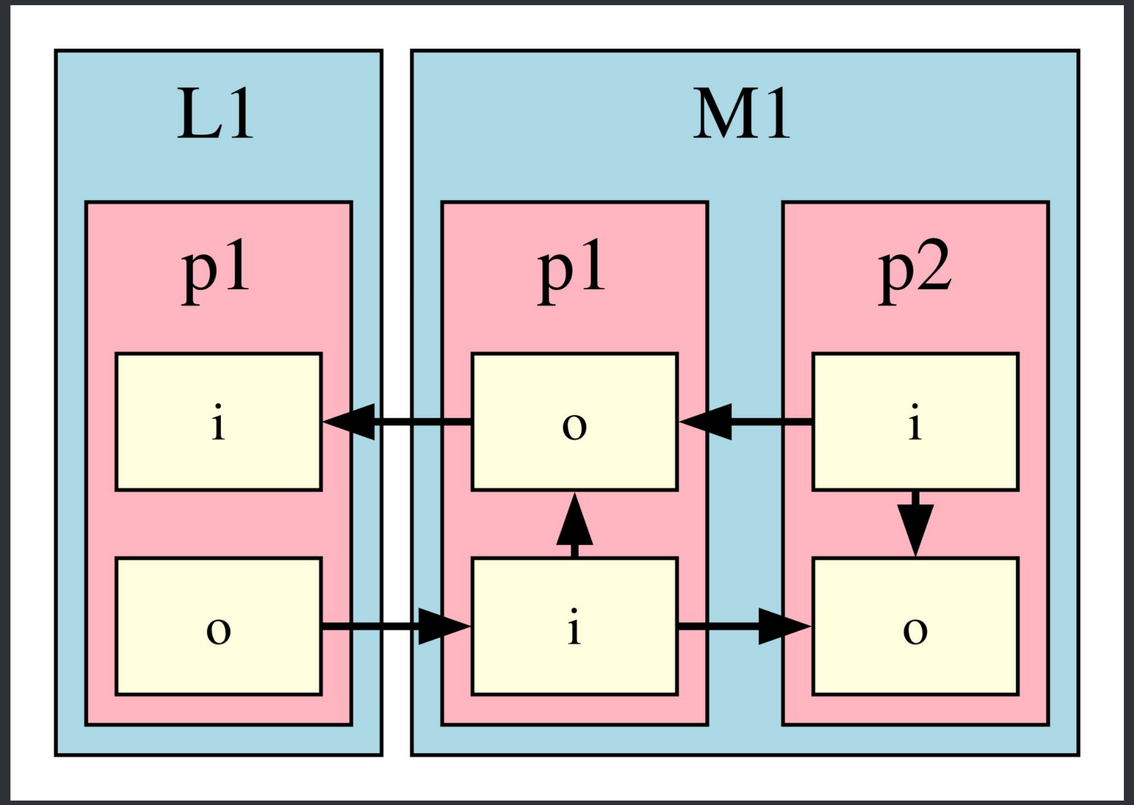
- Space length, index of refraction
- Mirror RTL coefficients, tuning

# Question from Jeremie



# Question from Jeremie: “What is a component”

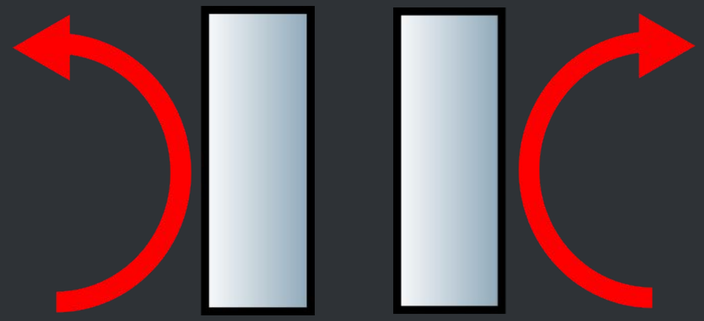
- You need to decide on your connection points
- You need to provide an equation for every black arrow
- You provide *parameters* that are used in the coupling equations
- Finesse can vary these parameters in a simulation



# 4. Filling the matrix







```
phi = pre_factor * freq.f

for field_idx in range(homs_view.shape[0]):
    n = homs_view[field_idx][0]
    m = homs_view[field_idx][1]

    if zero_tem00_gouy:
        ni = n
        mi = m
    else:
        ni = n + 0.5
        mi = m + 0.5

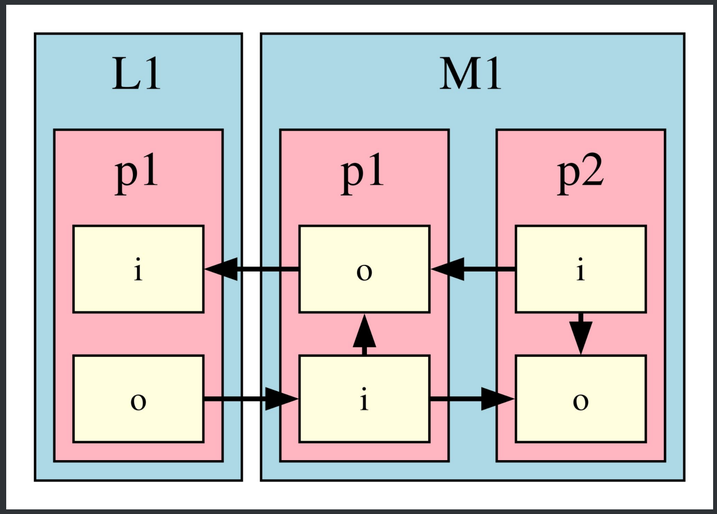
    gouy = ni * gouy_x + mi * gouy_y
    ws.couplings[field_idx] = cexp(1j * (-phi + gouy))

# Fill diagonals with the vector
if conn.P1i_P2o[freq.index]:
    (<SubCCSView>conn.P1i_P2o[freq.index]).fill_negative_zd_2(ws.couplings, 1)
if conn.P2i_P1o[freq.index]:
    (<SubCCSView>conn.P2i_P1o[freq.index]).fill_negative_zd_2(ws.couplings, 1)
```

```
# Reflections
if conn.P1i_P1o[freq.index]:
    (<SubCCSView>conn.P1i_P1o[freq.index]).fill_negative_za_zm_2(_r1, &ws.K11.mtx)
if conn.P2i_P2o[freq.index]:
    (<SubCCSView>conn.P2i_P2o[freq.index]).fill_negative_za_zm_2(_r2, &ws.K22.mtx)
# Transmission
if conn.P1i_P2o[freq.index]:
    (<SubCCSView>conn.P1i_P2o[freq.index]).fill_negative_za_zm_2(_t0, &ws.K12.mtx)
if conn.P2i_P1o[freq.index]:
    (<SubCCSView>conn.P2i_P1o[freq.index]).fill_negative_za_zm_2(_t0, &ws.K21.mtx)
```

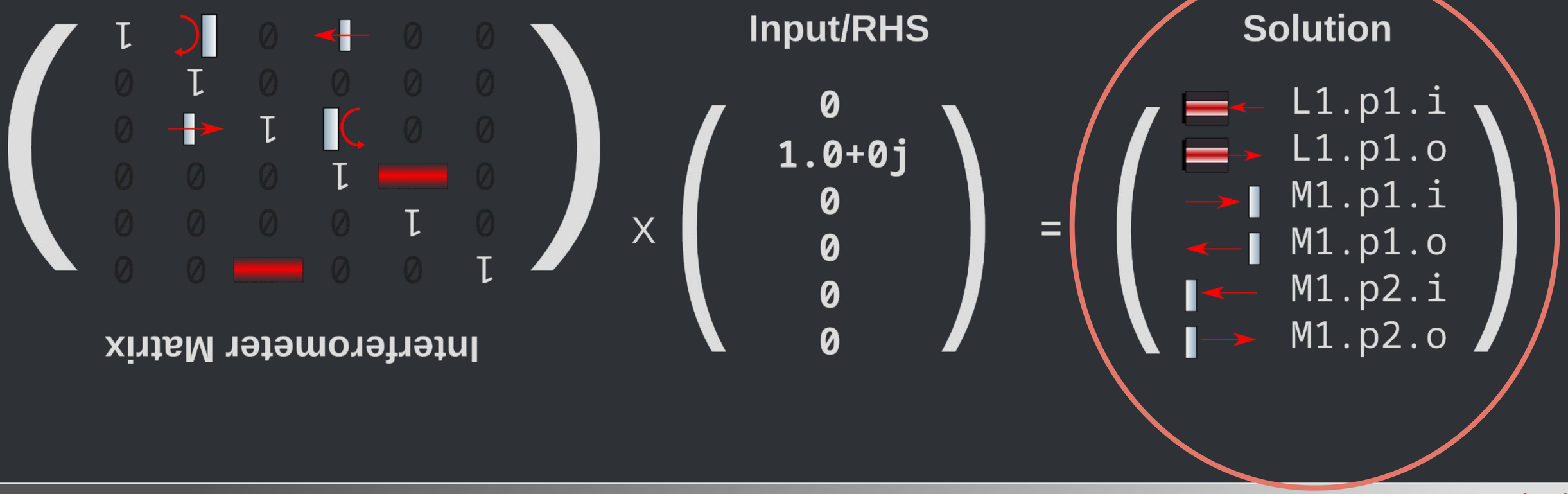
# 4. Filling the matrix

```
for component in model.components:
    for connection in component.connections:
        matrix.fill_coupling(connection, component)
```



# 5. Invert the Matrix

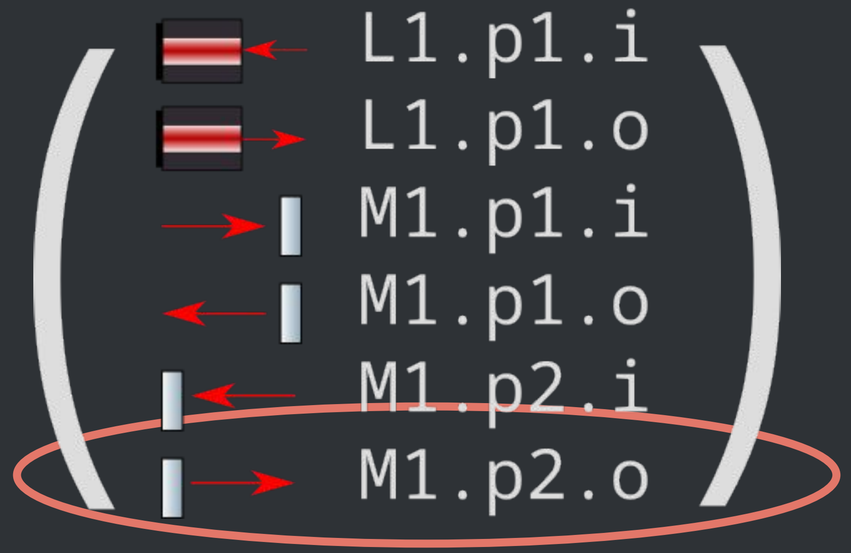


# 6. Calculate solution

```
laser L1 P=1
space S1 L1.p1 M1.p1
mirror M1 R=0.9 T=0.1
pd transmitted_power M1.p2.o
```

**Solution**




$$P = EE^*$$

```
•[1]: import finesse

      model = finesse.Model(
          """
          #kat
          laser L1 P=1
          space S1 L1.p1 M1.p1
          mirror M1 R=0.9 T=0.1
          pd transmitted_power M1.p2.o
          """
      )
      sol = model.run()
      sol["transmitted_power"]

 [1]: np.float64(0.10000000000000002)
```

# 6. Calculate solution

Xaxis action ⇒ ArraySolution

- Sweep Array
- Array of values for every detector
- Leaves the model in the original state

Maximize action ⇒ Optimize Solution

- Leaves the detector in a different state!
- The final value for all optimization parameters
- SciPy optimization info

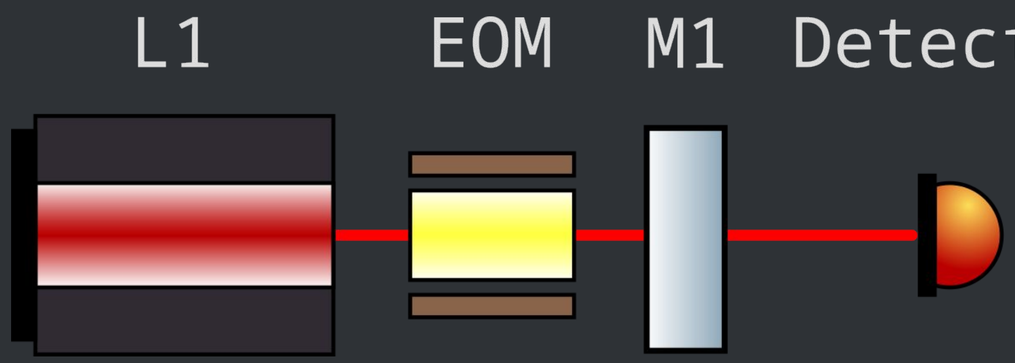
# Some examples in detail

- Plane wave
- **Modulation frequencies**
- Higher-order modes
- Transfer function / signal modelling

# Why add frequencies?

- Detectors use multiple optical sidebands to control the interferometer
- So we need to model these effects…

# Multiple Frequencies




```
laser L1 P=1
space S1 L1.p1 EOM.p1
modulator EOM f=1e6 order=1 midx=0.1
space S2 EOM.p2 M1.p1
mirror M1 R=0.9 T=0.1
pd transmitted_power M1.p2.o
```

We now have three fields for every node!

**Solution**



# Multiple Frequencies

- 3 Frequencies ⇒ 9 possible couplings between every node
- What does this mean for your coupling equations
- Different phases on reflection



# Multiple Frequencies

```
laser L1 P=1
space S1 L1.p1 M1.p1
mirror M1 R=0.9 T=0.1
pd transmitted_power M1.p2.o
```

**Solution**




$$P = EE^*$$

- Power detector: sums over frequencies
- Amplitude detector: select a frequency!
- Demodulated power detector

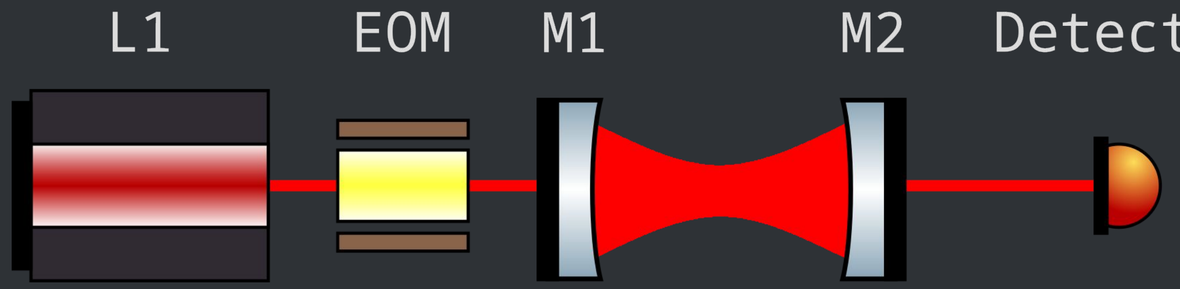
# Some examples in detail

- Plane wave
- Modulation frequencies
- **Higher-order modes**
- Transfer function / signal modelling

# Why add Higher-Order Modes?

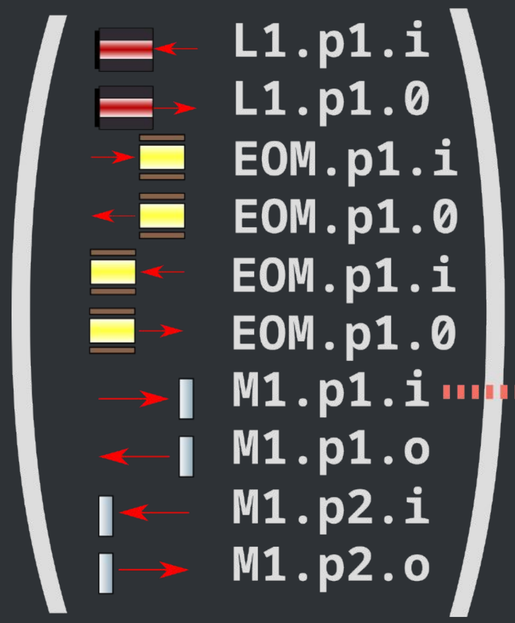
- Modelling the transverse shape of the beam
- Mirror surface imperfections
- Mirror misalignment

# Higher-Order Modes




9 fields at every node

Solution






# Higher-Order Modes

- 3 frequencies mean 9 couplings at every node connection
- 3 Higher Order modes means every coupling is a 3x3 scattering matrix ***K***

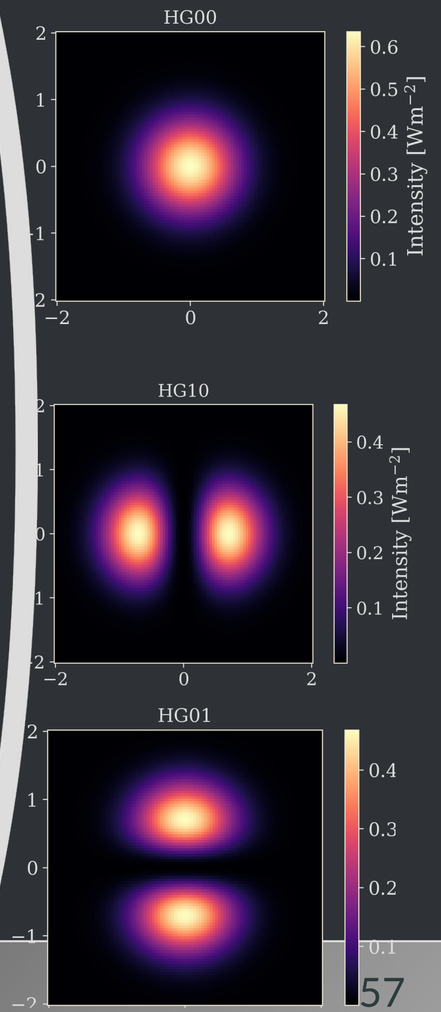


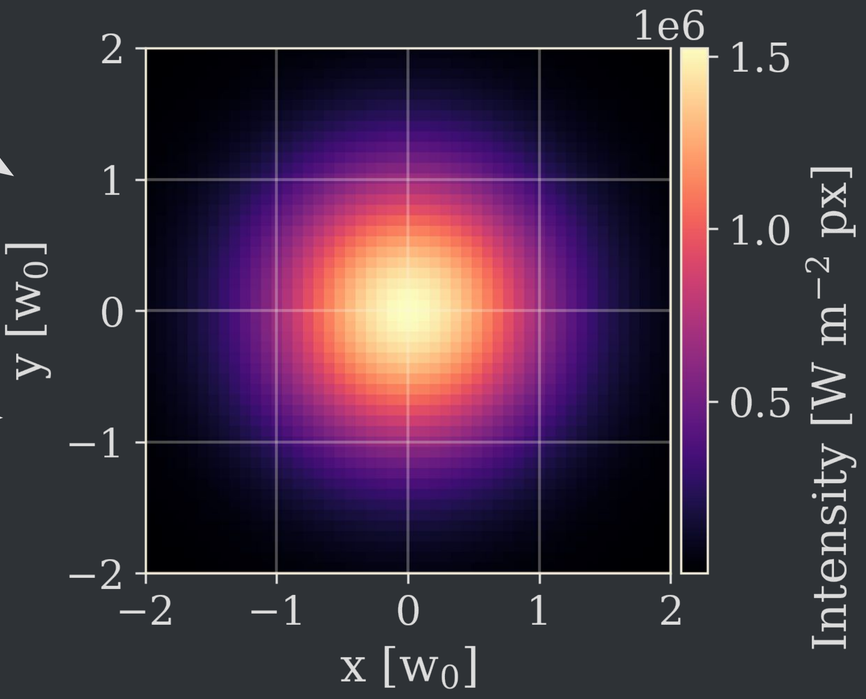
# Higher-Order Modes:

- Power detector sums over 9 fields
- Amplitude detector selects frequency and HOM (n,m)
- CCD detector calculates image based on HOM amplitudes

**Solution**

```
M1.p1.o f0 HG00
M1.p1.o f0 HG01
M1.p1.o f0 HG10
M1.p1.o f1 HG00
M1.p1.o f1 HG01
M1.p1.o f1 HG10
M1.p1.o f2 HG00
M1.p1.o f2 HG01
M1.p1.o f2 HG10
```



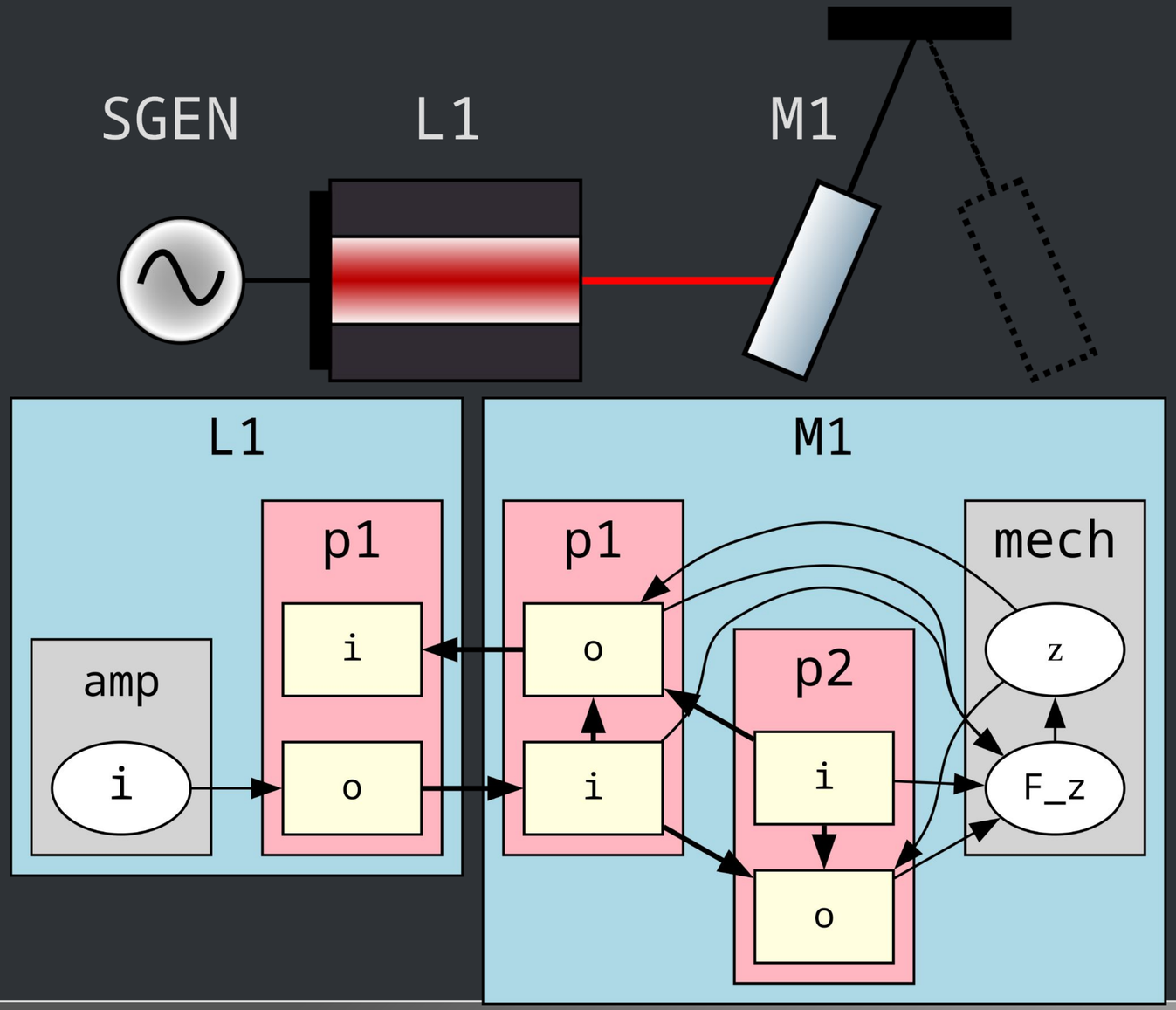
# Which examples will I show

- Plane wave
- Modulation frequencies
- Higher-order modes
- **Transfer function / signal modelling**

# What is signal modelling?

“One of the primary tasks undertaken with Finesse is simulating the frequency domain behaviour of complex optical systems. This involves computing various transfer functions of small signals injected into the system and computing how they affect various outputs. With knowledge of how these small signals behave in an optical system we can then model how classical and quantum noise propagates throughout an optical system and the behaviour of control systems designed to keep the an optical system operating.”

# Signal modelling




```
import finesse

model = finesse.Model(
    """
    #kat
    laser L1 P=1
    space s1 L1.p1 M1.p1
    mirror M1 R=1 T=0

    pendulum suspension M1.mech mass=1 fz=1000 Qz=1000
    sgen amp_signal L1.amp.i
    motion_detector M1_z M1.mech.z
    """
)
```

# Signal modelling

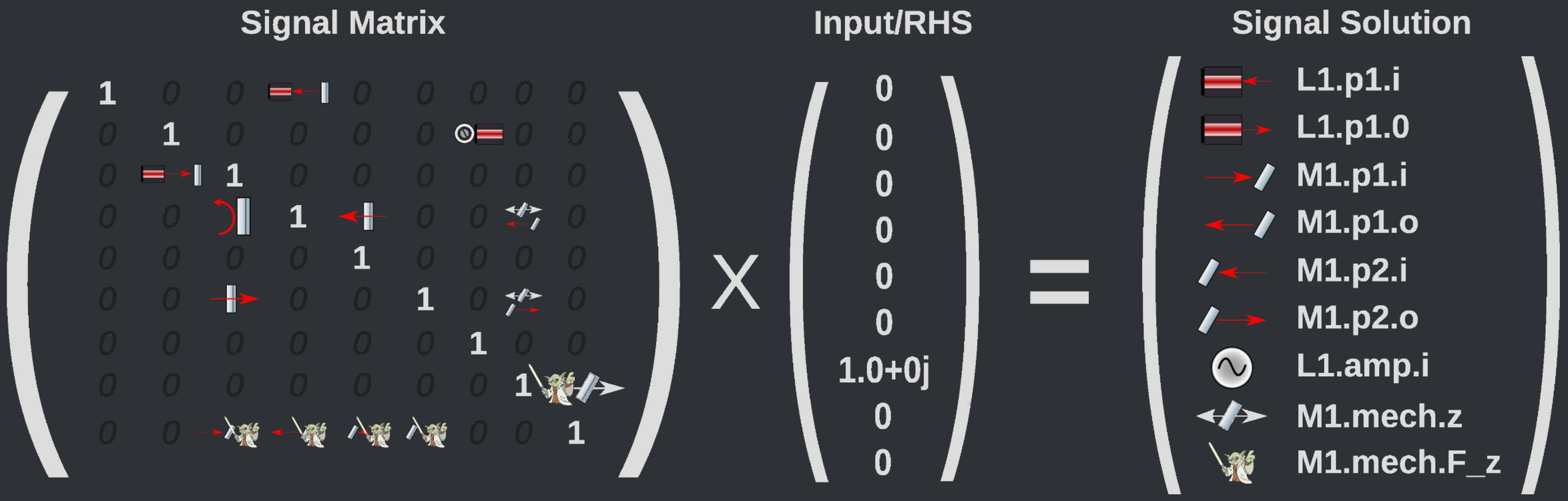
## Carrier Solution




## Signal Solution



# Signal modelling



# Signal modelling



- Signal injection coupling



- Force2position: suspension physics, frequency dependent

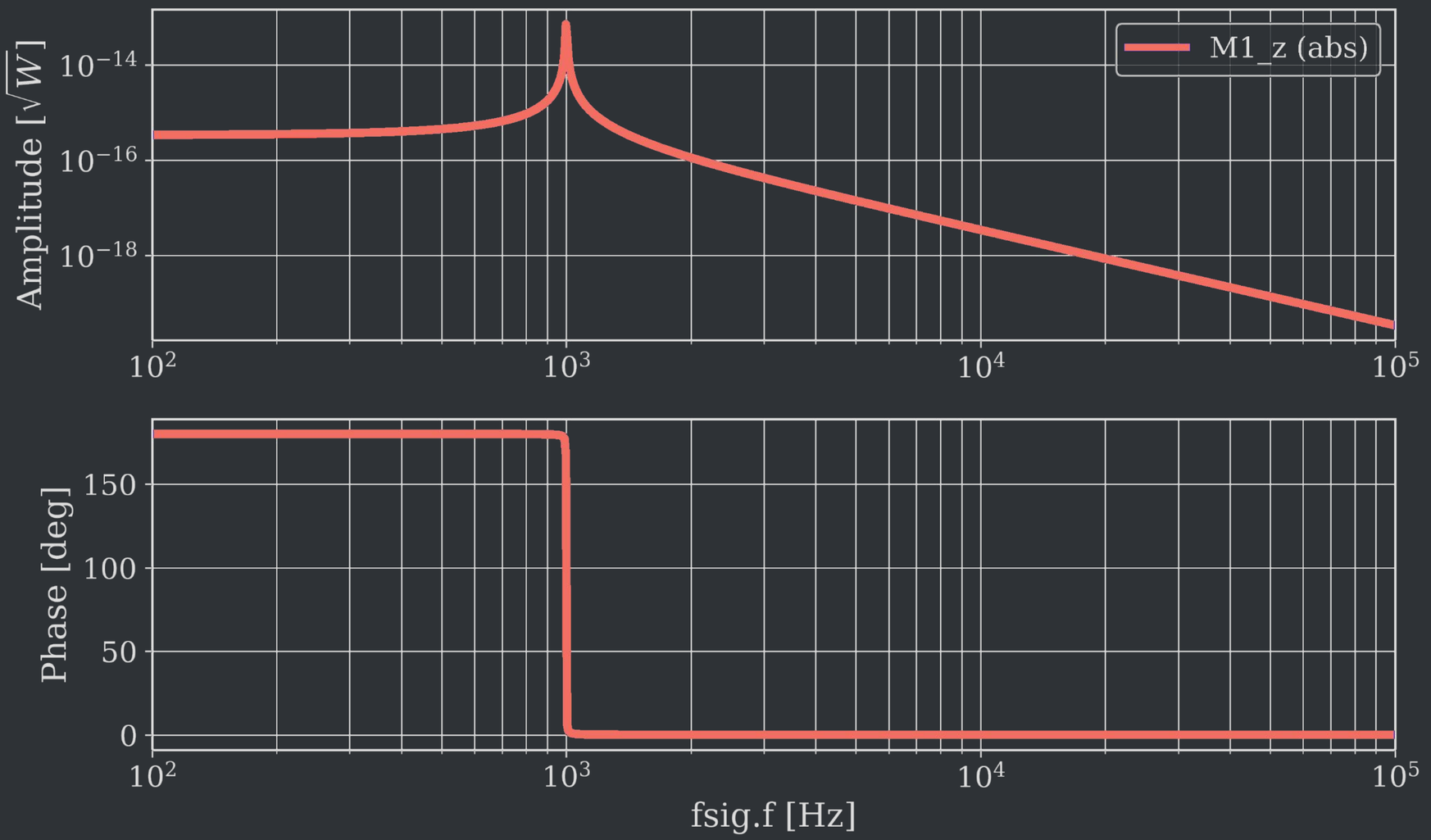


- Field to force coupling: Radiation pressure

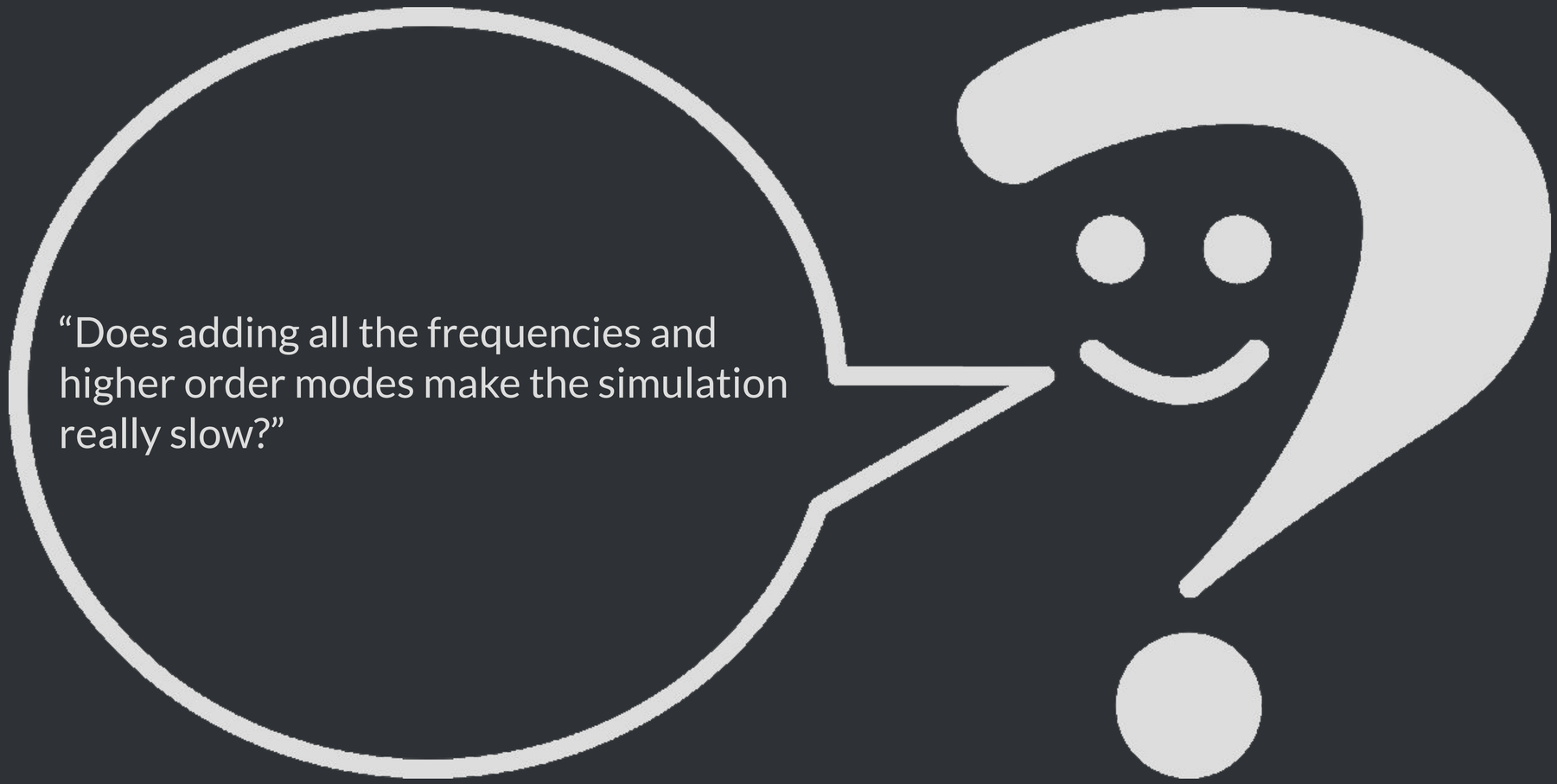


- Position to outgoing field: reflection

# Signal output

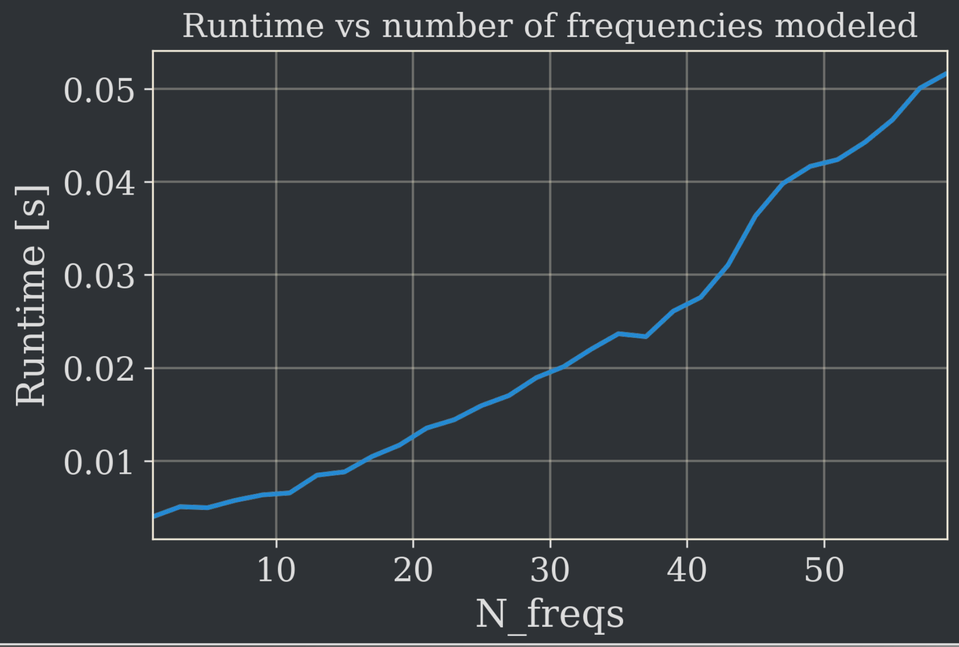


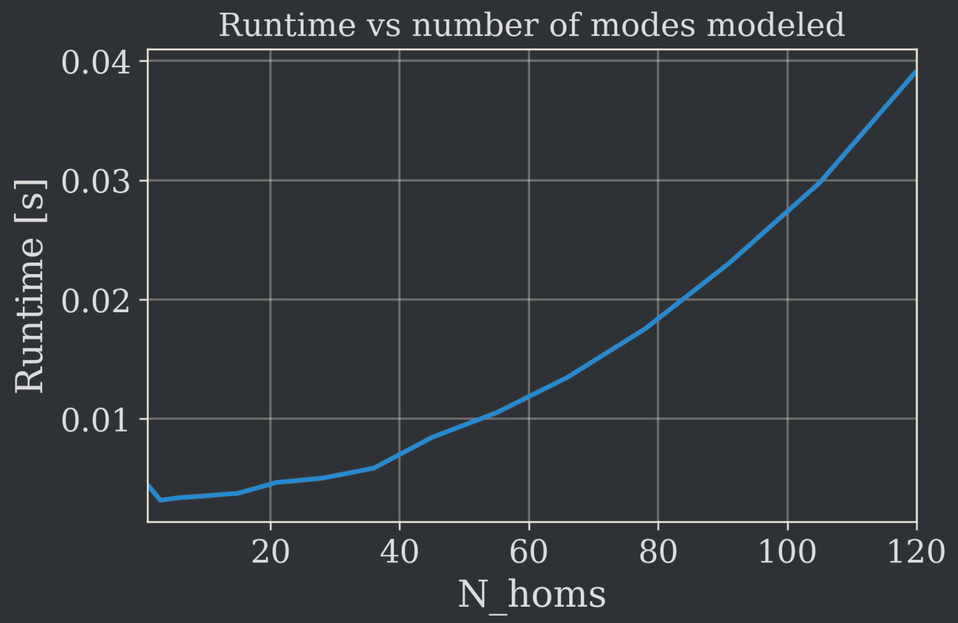
# Question from Jeremie



# Cost of complexities

1. Multiple frequencies → Modulator ⇒ Matrix size: N_nodes * N_freqs
2. HOMs ⇒ cavity/beam shape ⇒ Matrix size: N_nodes * N_freqs * N_homs
3. Signal modelling (coherent noise) ⇒ Add a second matrix with more nodes



# Seeing the Matrix

```python
import finesse

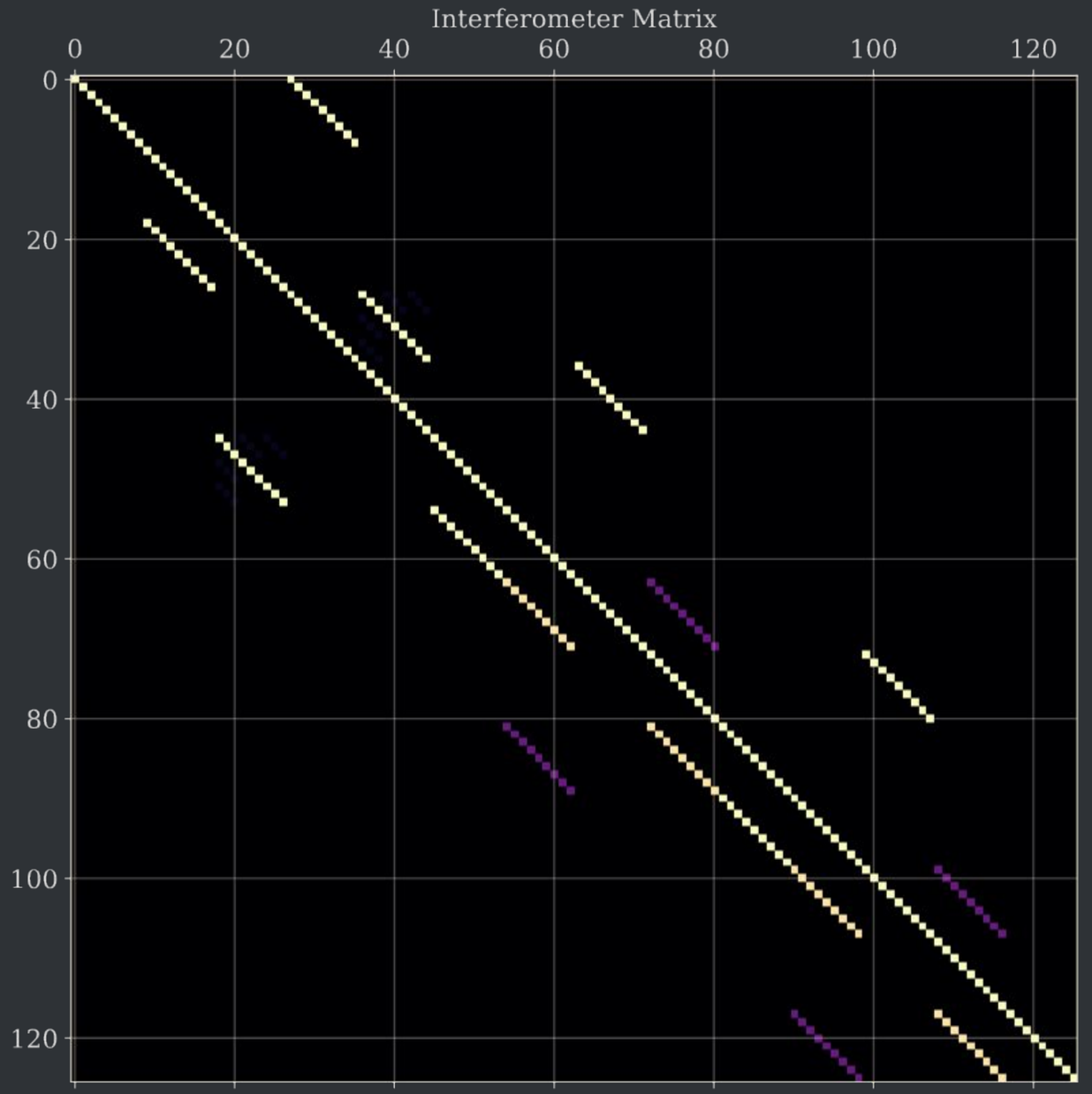
model = finesse.Model(
    """
    #kat
    laser L1 P=1
    space S1 L1.p1 EOM.p1
    modulator EOM f=1e6 midx=0.1 order=1
    space S2 EOM.p2 M1.p1
    mirror M1 R=0.9 T=0.1 Rc=-9
    space s_cav M1.p2 M2.p1 L=10
    mirror M2 R=0.9 T=0.1 Rc=9
    cav cavity M1.p2.o
    modes(maxtem=1)
    """
)
model.run(simulation_options={"debug_mode": True})
```

```
Matrix carrier: nnz=522 neqs=126
    (col, row) = value
    (0, 0)     = (1+0j)   : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=0 -> I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=0
    (1, 1)     = (1+0j)   : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=1 -> I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=1
    (2, 2)     = (1+0j)   : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=2 -> I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=2
    (3, 3)     = (1+0j)   : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=0 -> I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=0
    (4, 4)     = (1+0j)   : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=1 -> I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=1
    (5, 5)     = (1+0j)   : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=2 -> I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=2
    (6, 6)     = (1+0j)   : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=0 -> I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=0
    (7, 7)     = (1+0j)   : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=1 -> I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=1
    (8, 8)     = (1+0j)   : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=2 -> I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=2
    (9, 9)     = (1+0j)   : I,node=L1.p1.o,f=0,fidx=3,Neq=3 mode=0 -> I,node=L1.p1.o,f=0,fidx=3,Neq=3 mode=0
    (9, 18)    = (-1-0j)  : I,node=L1.p1.o,f=0,fidx=3,Neq=3 mode=0 -> I,node=EOM.p1.i,f=0,fidx=6,Neq=3 mode=0
```

```
Vector carrier: neqs=126
    (row) = value
    (0)   = (-0.006637269507885192 6+5.928347363281739e-15j) : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=0
    (1)   = 0j                                               : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=1
    (2)   = 0j                                               : I,node=L1.p1.i,f=0,fidx=0,Neq=3 mode=2
    (3)   = (0.01640243670493658+0.06628969461928456j)       : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=0
    (4)   = 0j                                               : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=1
    (5)   = 0j                                               : I,node=L1.p1.i,f=1,fidx=1,Neq=3 mode=2
    (6)   = (-0.016402436704937177+0.06628969461928458j)     : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=0
    (7)   = 0j                                               : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=1
    (8)   = 0j                                               : I,node=L1.p1.i,f=2,fidx=2,Neq=3 mode=2
```

# Seeing the Matrix

- Non-zero entries of the matrix
- Brighter color is stronger coupling
- Black is no coupling
- 9 fields, groups of 9, 18 & 27
- Most of the coupling is very local!



# Thank you for your attention

Enjoy finessing!

Antonella will give a more basic finesse introduction and introduce the workshop projects.